

MALMÖ UNIVERSITY

Bridging the Skill Gap: The Role of AI in Upskilling and Reskilling the Tech Industry

A Qualitative study on the impact of AI in the tech workforce development in Colombia

Juan David Ramirez Jaramillo

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Supervisor: Dr. Katarzyna Gruszka

Popular Science Summary

Consider a world where learning is as personal and dynamic as your Spotify playlist, tailored to your objectives, calibrated to your pace, and accessible when you need it. That's the promise of Generative Artificial Intelligence (GenAI) in the field of educational technology. But has this been materialised in actual workplaces? More importantly, how can we use this technology to make learning more accessible, inclusive, and impactful?

This thesis examines the extent to which GenAI is being adopted or has the potential to be adopted to assist professionals in Colombia's technology sector in acquiring new skills and remaining competitive in an evolving market. To address these questions, I spoke with employees from tech companies, educators, and HR executives to gather insights on the use of GenAI for training and talent development purposes.

The analysis found that although some companies have begun to experiment with Genai, for instance, by developing learning pathways and recommending educational materials, the majority still operate without a defined strategy or vision for the future. Many grapple with practical challenges such as insufficient resources, infrastructure, time, or intra-organizational collaboration. Curiosity exists, but it is tempered with caution; there is a strong motivation to use AI, but only if it is employed wisely.

From these findings, I developed a set of practical guidelines that organisations can use to implement GenAI responsibly, focusing on purpose and ethics while emphasising authentic educational outcomes for professionals.

Ultimately, this research suggests that Genai could help bridge skill gaps, promote democratisation, and foster social inclusion, all with proper precautions. In Colombia, the focus is not solely on catching up with global trends; it’s about building more intelligent, fairer systems for the future of work.

Abstract

This study aims to investigate the integration of Generative Artificial Intelligence (GenAI) within workforce development programs in IT companies and higher education institutions in Colombia, with a special focus on addressing the country's skills gaps and reducing unemployment among young and disadvantaged populations. Given Colombia's status as an emerging economy, distinct challenges such as infrastructure limitations, organisational readiness, regulatory constraints, and digital literacy deficits will be closely analysed, alongside the unique opportunities GenAI provides, including personalised learning paths, scalability, and enhanced workforce readiness.

Using qualitative research methodologies, including semi-structured interviews with key Colombian stakeholders, as well as a comprehensive analysis of documents, reports, and public policies, this study aims to provide an in-depth understanding of stakeholder perceptions regarding the effectiveness, scalability, and ethical considerations of GenAI-based reskilling and upskilling initiatives. Ultimately, the research aims to provide strategic recommendations and an insightful analysis for the sustainable and inclusive implementation of GenAI-based training and learning solutions in Colombia's tech industry, thereby contributing to both workforce development and broader socio-economic inclusion.

Keywords— Generative Artificial Intelligence, workforce development, AI adoption, reskilling, upskilling, Learning and Development, Colombia

Declaration of AI Use

During the development of this thesis, I used various AI-based tools to support and enhance different stages of the research and writing process. These tools were employed thoughtfully and ethically, following academic standards and with full awareness of their capabilities and limitations.

Tools Used:

- Generative AI tools: ChatGPT (OpenAI), Gemini (Google), and Perplexity.
- Language enhancement tool: Grammarly.
- Document analysis tool: Adobe Acrobat AI Assistant.

Purposes:

- Idea development and refinement: Chatgpt, Gemini, and Perplexity were utilised during the early and intermediate stages to brainstorm research angles, compare perspectives, explore potential arguments, and refine titles and subtitles.
- Text support and clarification: These generative tools also assisted in rephrasing paragraphs for clarity, summarising content, and translating interview fragments originally conducted in Spanish into English. At times, they were used to locate relevant quotes within coded transcripts and analyse connections within the data more efficiently.
- Language correction: Grammarly was used to improve grammar, syntax, and fluency across the final document, particularly in ensuring clarity for an international academic audience.
- Literature review navigation: Adobe Acrobat's AI Assistant was used to more efficiently search, highlight, and interpret academic articles and policy reports. This tool helped locate specific references, extract citations, and deepen analysis without relying solely on manual scanning.

Scope and Ethical Considerations:

The AI tools used did not replace critical thinking, analytical judgment, or the author's original contributions. They were applied as supporting resources to enhance efficiency and linguistic precision; however, all research design, thematic analysis, and data interpretation were independently conducted by the author.

Additionally, the design and content of the final practical guidelines (the thesis artefact) are the author’s original creation, fully grounded in the empirical findings and conceptual insights developed throughout the research process.

My use of AI tools was in line to proper academic conduct, ensuring transparency, originality, and accountability throughout the research process.

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1 Introduction

In recent decades, technological advances have had a significant impact on various industries, including banking, e-commerce, education, and industrial production. Automation and the implementation of artificial intelligence have radically transformed how we conduct our tasks, presenting the potential to add trillions of dollars to the global economy. According to PwC(PwC, 2024) estimates, artificial intelligence is expected to contribute an increase of up to $15.7 trillion to the global economy by 2030. Additionally, McKinsey & Company(Chui et al., n.d.) argues that generative artificial intelligence could increase AI's impact by 40%, adding between $2.6 and $4.4 trillion.

There are various perspectives on the potential job destruction resulting from these advances. A comprehensive study by Carl Benedikt Frey and Michael Osborne(Frey & Osborne, 2017), researchers at the Oxford Martin School, projects that 47% of jobs could disappear by 2033 due to automation. Furthermore, Nguyen(Nguyen et al., 2020) warns that disproportionate technological progress compared to productive workforce additions could increase unemployment. This job reduction would primarily affect workers with lower skills, who would be more likely to be replaced by machines.

However, technological advances are expected to have a positive impact on the job market in various industries in the next five years. According to the World Economic Forum(*The Future of Jobs Report 2023*, n.d.), technology is likely to generate at least 12 million more jobs than it destroys by 2025.

The current landscape of the global job market is characterised by a significant skills deficit, as evidenced by a study conducted by Amazon. This report reveals that 78% of a sample of 3,000 employees perceive a lack of essential skills for the future(Dan Schawbel, n.d.). This skills deficit hampers companies from filling crucial positions, placing them at a disadvantage in a highly competitive market such as digital services and products. According to Nash Squared's Digital Leadership Report 2022, which involved 82 countries, 58% of IT business leaders anticipate an increase in demand for specific skills, while 70% perceive that the shortage of these skills is hindering their growth projection(Colback, 2023).

In this scenario, Generative Artificial Intelligence appears to play a central role. The tech revolution is reshaping our work landscape in ways we couldn't have imagined just a few years back. LinkedIn's latest Future of Work Report(Linkedin, 2023) reveals that cognitive technology workers are experiencing the most pronounced disruption, a predictable outcome, with retail sector professionals encountering equivalent transformation rates, both positioned at approximately 71%. In close proximity are wholesale commerce (68%), financial services (66%), and specialised professional services (64%).

The compelling aspect of this phenomenon extends beyond a statistical representation to its implications for career development and mobility. As organisations across diverse sectors rapidly implement these advanced tools, proficiency in artificial intelligence transcends conventional skill enhancement to become a form of professional currency, enabling talented

individuals to traverse previously impermeable industry boundaries with remarkable efficiency. We will see professionals with solid AI knowledge transition from healthcare to fintech without missing a beat, something that would have required years of retraining before, but not today.

Generative AI is presented as a solution to this problem. This technology offers multiple benefits to improve the learning of these skills, including personalisation of the educational process, adaptation to the specific needs of each student, more private and practical support, adaptive feedback, and interactive learning (Imram & Torralba, 2024). Additionally, GenAI tools enhance motivation and participation, as learning through a conversational agent energises the educational process and increases student engagement (Okonkwo & Ade-Ibijola, 2021).

In the context of Colombia, leveraging Generative Artificial Intelligence (GenAI) presents a significant opportunity to address the persistent skills gap, reduce youth unemployment, and promote greater economic inclusion and competitiveness within the tech industry. Nevertheless, adopting such advanced technological solutions poses unique challenges, including infrastructural limitations, digital literacy gaps, and regulatory constraints that must be understood and addressed strategically. This research aims to explore these dynamics, providing actionable insights deeply and targeted recommendations for effectively integrating GenAI into Colombia's workforce development strategies, ultimately driving sustainable, inclusive, and equitable growth within the nation's rapidly evolving technological landscape.

1.1 Key Terminology

- **Artificial Intelligence (AI):** Artificial intelligence (AI) is a technology that enables computers and machines to simulate human learning, comprehension, problem-solving, decision-making, creativity, and autonomy. (Stryker, 2024)

- **Generative AI (GenAI):** A subset of AI. It can be thought of as a machine-learning model trained to create new data, rather than predicting a specific dataset.(Zewe, 2023) It includes technologies such as ChatGPT, DALL·E, and AI-driven content generators.

- **Upskilling:** According to LinkedIn, it refers to acquiring new skills or enhancing existing skills to stay competitive in the job market. Upskilling is specifically focused on obtaining knowledge, expertise, or capabilities related to your current field or industry to advance your career or adapt to changes in the job market (LinkedIn, n.d.).

- **Reskilling:** It refers to the acquisition of entirely new skills. Reskilling is often pursued to pivot to a different career or industry due to changes in job demand or personal career goals (Stanford Engineering Center for Global & Online Education, 2023).

- **Workforce Development:** The OECD defines it as a labour market strategy that includes employment and training services. It involves coordinating public and private

sector activities, policies, and programmes to create and sustain a workforce that can support current and future business and industry. (OECD, 2020)

1.2 Research Problem, Purpose and Objectives

The research problem centers on the challenges and opportunities associated with integrating Generative Artificial Intelligence (GAI) into workforce development in the tech industry within IT companies and higher education institutions in Colombia, specifically to address persistent skill gaps and youth unemployment in that sector. Despite the significant potential of GAI to facilitate personalized and scalable reskilling and upskilling solutions, Colombia faces distinct barriers such as infrastructural deficits, limited digital literacy, regulatory constraints, and socio-economic disparities. Consequently, current workforce training and education strategies struggle to match the accelerating pace of technological progress, creating a more significant gap between industry demands and the available skilled talent.

This study aims to investigate the key challenges and opportunities associated with implementing Generative Artificial Intelligence (GenAI) to support upskilling and reskilling initiatives within the tech industry and higher education institutions in Colombia. The research specifically aims to provide actionable insights and recommendations for integrating GenAI solutions to effectively reduce skill gaps, enhance workforce readiness, and improve economic inclusion, focusing on the needs and realities of the Colombian context.

In order to achieve that goal, the following specific objectives are suggested:

1. Asses the current state of GenAI adoption and implementation in workforce training programs in IT companies and higher education institutions in Colombia.
2. Identify barriers and enablers influencing the successful integration of GenAI for upskilling and reskilling initiatives.
3. Evaluate and analyse the opinions of key stakeholders regarding efficacy, scalability, and ethical considerations on AI-driven learning, training, and development strategies.
4. Develop a set of guidelines that includes the main insights and strategic recommendations to optimise the use of GenAI in workforce development for IT.

1.3 Research Questions

To achieve the research goals, the study is guided by the following key questions:

1. What are the challenges and opportunities of implementing GenAI in upskilling and reskilling programs for tech professionals in Colombia?
    a. This question aims to identify the barriers hindering AI adoption in labour force training, such as infrastructure constraints, regulatory issues, and organisational resistance, and to highlight the benefits of AI-based learning, including its ability to personalise training for individuals, its scalability, and affordability.

b. It is expected to answer this question through semi-structured interviews and surveys with key stakeholders ranging from education representatives to IT industry professionals, as well as a review of existing AI training initiatives in both countries.

2. What are the perceptions of key stakeholders in Colombia (industry leaders, professionals, educators) regarding the efficacy, scalability, and ethical implications of using Generative AI in reskilling and upskilling initiatives?
    a. This question aims to reveal stakeholders' insights into GenAI´s potential effectiveness, fairness, and practicality, ensuring recommendations are responsive to local values, cultural contexts, and socio-economic realities.
    b. To answer this question, the study will conduct a qualitative analysis based on information gathered from the interviews, and the analysis of grey literature on AI implementation in workforce training, identifying patterns and best practices along with contextual differences.

3. How can we utilise GenAI to bridge skill gaps in the tech workforce, particularly in emerging economies like Colombia?
    a. The aim here is to uncover actionable strategies that utilise GenAI not only as a technological solution but also as a mechanism for broader socio-economic development.
    b. After conducting the qualitative analysis of the interviews, a guideline will be synthesised to inform L&D and educators about AI-enable workforce training , providing practical, scalable, and context-sensitive insights that will lead to the creation of valuable solutions within the organisation

1.4 Relevance and Contribution to Sustainability

This research makes a significant contribution to the United Nations' 2030 Agenda for Sustainable Development by exploring how Artificial Intelligence can be utilised to develop more advanced, inclusive, innovative, and scalable workforce development strategies. By examining how AI and GAI can support economic inclusion, digital transformation, and educational innovation in emerging economies, using the Colombian context as a case study.

At its core, the research aligns with three key Sustainable Development Goals:

- **SDG 4 – Quality Education**
- **SDG 8 – Decent Work and Economic Growth**
- **SDG 9 – Industry, Innovation, and Infrastructure**

The table below illustrates how this research aligns with specific targets and indicators for each of the Sustainable Development Goals (SDGS) mentioned, highlighting our concrete contributions.

Target	SDG Target & Indicator	Research Contribution
T4.3 Access to Affordable Education	***Indicator 4.3.1:***	Expands access to non-formal, on-demand GenAI training for tech professionals and adult learners. Reduces barriers to learning by providing flexible, digital pathways, especially relevant for under-resourced regions.
Ensure equal access to technical, vocational, and tertiary education.	Participation rate in formal and non-formal education and training.	
T4.4: Relevant Skills for Employment	***Indicator 4.4.1:***	Promotes AI-driven digital skills training and upskilling through GenAI, tailored to industry needs. Enables tech professionals and unemployed individuals to develop ICT and soft skills essential for employment in digital economies.
Increase the number of youth and adults with technical and vocational skills.	Proportion of youth and adults with ICT skills, by type of skill.	
T8.6: Youth Employment	**Indicator 8.6.1:**	Targets NEET reduction in Colombia through accessible GenAI tools for self-paced and low-cost learning. Enables re-engagement in the labor market via short-term, scalable learning paths adapted to disadvantaged youth.
Reduce the number of young people disconnected from learning or work.	Proportion of youth (15–24) not in education, employment or training (NEET).	
T8.5 Full and Productive Employment	**Indicator 8.5.1:**	Supports inclusive learning ecosystems that upskill underrepresented groups (e.g., women, young people, low-income individuals), enhancing their access to better jobs
Promote decent work and equal pay for all.	Average earnings by sex, age, occupation, and disability status.	
T9.5: Research and Innovation Capacity	**Indicators 9.5.1 & 9.5.2:**	Encourages research-based adoption of AI in education and workforce development. Promotes public-private-academic partnerships that build institutional capacity in Colombia to implement and scale GenAI ethically and sustainably.
Enhance technological capabilities and innovation, especially in developing countries.	R&D spending	
	Number of researchers.	

Table 1. SDG Research contribution matrix

SDG 4: Quality Education

The research directly supports SDG 4 by investigating how GenAI can foster personalised and accessible learning pathways in corporate and academic settings. AI can help people learn in a more innovative and personalised way, almost like having a private tutor on their phone or computer(Pavitra & Agnihotri, 2023; Storey & Wagner, 2024). This is especially helpful for people in Colombia who may not have access to good training. By using AI, they can still get a quality education, tailored to what they need to learn to find a job.

Moreover, the study emphasises the democratisation of learning through digital tools that reduce barriers related to geography, infrastructure, or economic status. This is particularly relevant for Colombia, where educational inequality and digital exclusion persist. As stated in the proposal, this research promotes AI-led learning models for accessibility and personalisation relevant to workforce training.

SDG 8: Decent Work and Economic Growth

The study supports SDG 8 by targeting the gap between existing skills and job market needs. Generative AI is positioned as a tool that enhances employability and supports career transitions, enabling workers to remain competitive in the face of automation and digital transformation(Sidhu et al., 2024). In Colombia, where youth unemployment and job informality are pressing concerns, GenAI offers an affordable and scalable way to deliver vital skills to underserved communities.

The study also explores how AI can shift HR professionals into more strategic roles by automating repetitive tasks and redirecting focus toward continuous learning, employee development, and job satisfaction.

SDG 9: Industry, Innovation and Infrastructure

AI is a big part of the future, but not every country is equally prepared. Regarding SDG 9, the research demonstrates how GenAI can enhance digital infrastructure and fuel innovation in workforce training. Through an analysis of implementation challenges in Colombia, the study addresses the infrastructural and regulatory limitations that hinder the adoption of AI in developing contexts.

This research bridges technological innovation with social inclusion by leveraging Generative AI to support education, employment, and infrastructure development in diverse economic contexts. The insights and strategic recommendations developed through this study will contribute to the global discourse on harnessing AI to reduce inequality, promote economic resilience, and accelerate progress toward the 2030 Agenda. This research empowers stakeholders to align AI implementation with sustainability goals and long-term societal well-being by supporting IT companies and higher education institutions with actionable knowledge.

2 Background and Related Work

2.1 The Global Skills Gap and Workforce Disruption

The rapid technological progress in global labour markets has created increasing worries about how workers match their skills to available jobs and their readiness for future work. The Work Change Report 2025 (Linkedin, 2025) predicts that 70% of current job skills will transform by 2030 due to artificial intelligence (AI). The modern workforce requires employees to switch careers at least twice during their working lives, according to LinkedIn's 2025 projections.

McKinsey & Company(Kweilin Ellingrud et al., 2023) predicts that 30% of present work activities will become automated by 2030, specifically in positions that perform routine physical or cognitive tasks. The changes in the job market affect both blue-collar workers and white-collar professionals, creating an immediate need for new skills. The Harvard Business Review states that workforce adaptation has become an immediate requirement because Generative AI (GAI) is quickly integrating into operational workflows throughout all industries. (Jorge Tamayo et al., 2023)

The accelerating rate of technological advancements hinders the gradual development of educational systems, creating an essential skills deficit. Employers and employees must adopt a lifelong learning approach, incorporating AI-based upskilling and reskilling models, to remain competitive.

2.2 Technological Context: The Rise of AI and Gen AI in Workforce Development

The development of the workforce highly depends on Artificial Intelligence. Between 2015 and 2023, the hiring rate for AI-related roles increased by more than 300%, while AI-literate professionals expanded their presence across various industries. Generative AI tools, including Chatgpt and Microsoft Copilot, have accelerated this trend because they provide real-time content generation, personalised tutoring, and intelligent task automation capabilities. McKinsey reports that generative AI tools have become essential for more than 75% of knowledge workers who started using them at their workplaces last year.

GAI implementation in learning and development enables organisations to deliver training at scale while tailoring it to individual needs. Organisations implementing AI-based upskilling programs observe significant productivity growth, along with enhanced employee engagement(Eric Hazan et al., 2024). The real-time interaction capabilities of AI systems enable learners to develop human skills, such as critical thinking and communication, while acquiring technical skills.

The advantages of these benefits do not eliminate existing obstacles. The deployment of training systems based on AI or GAI faces challenges due to data privacy concerns and algorithmic bias, as well as unequal access to AI infrastructure, which particularly affects developing economies.

2.3 Related Work and Research Gap

A literature review (LR) has been conducted with the aim of gathering information that will allow me to discover the state of research on the application of generative artificial intelligence in adult education, especially its prevalence and applicability to facilitate and enhance skills development processes in technology companies. Special emphasis was placed on the Colombian context, given its unique challenges and opportunities in adopting these technologies. The complete LR process can be reviewed in Appendix A.

The incorporation of Generative AI (GenAI) into workforce development and adult education has been assessed extensively in the literature. There are studies in developed markets, like Sweden, that expose how GenAI could facilitate the scaling of technical skill development where the demand for high-tech expertise is on the rise. For example, Bridging the Skill Gap in IT by Using Generative AI(Tenger & Taeymans, 2024) shows that GenAI can enhance technical skill development through personalised learning journeys despite challenges such as data accuracy, algorithmic bias, and accessibility.

In emerging economies like Colombia, substantial barriers to AI adoption have been documented. Studies such as "AI and Workforce Development: A Comparative Analysis"

(Sidhu et al., 2024) and "Adopción de Tecnologías de Inteligencia Artificial en Colombia"(Otálvaro, n.d.) Illustrate significant obstacles, including infrastructure deficits, regulatory constraints, limited technical expertise, and financial constraints. These studies highlight the necessity for government interventions, targeted investments in digital infrastructure, and collaborative public-private efforts to facilitate broader AI implementation in workforce development. Despite these foundational efforts, practical GenAI implementations in Colombia remain limited, often hindered by the lack of scalable and sustainable training solutions.

Broader insights from "Artificial Intelligence in Corporate Learning and Development: Current Trends and Future Possibilities"(Pavitra & Agnihotri, 2023) show how corporate environments can leverage AI to optimise employee learning through personalised recommendations, skill-gap assessments, and automated learning content creation. These opportunities present valuable implications for Colombian corporations attempting to bridge local skill gaps using GenAI.

In adult education, some studies put focus on GenAI capabilities in creating a personalised learning path, automating content creation processes, and addressing resource gaps in education systems (Cacicio & Riggs, 2023). Additionally, "Integrating Artificial Intelligence (AI) into Adult Education: Opportunities, Challenges, and Future Directions"(Storey & Wagner, 2024) emphasises the critical role AI could play in adult education contexts similar to Colombia's, by enhancing flexibility, inclusivity, and accessibility of learning experiences, while also acknowledging challenges such as digital literacy gaps, data privacy, and ethical concerns. Furthermore, researchers from the University of Bologna(Morandini et al., 2023) underscore the transformative potential of AI-driven training in addressing the evolving demands of the digital economy, presenting evidence on how AI-enabled skill development interventions can significantly benefit organisations undergoing rapid technological changes.

These studies demonstrate the potential for GenAI to offer education with greater flexibility and accessibility in resource-constrained settings. However, they lack frameworks or guidelines for sector-specific applications, particularly in the technology sector, and this is where the dynamics of AI integration differ significantly from those in education in general.

Broader socio-economic analyses, such as those on Artificial Intelligence in Industry 4.0and Artificial Intelligence in Developing Countries(Mhlanga, 2021) address AI's potential to drive economic growth, reduce inequality, and create new job opportunities. While these explorations highlight the future relevance of AI as a guide, they mostly remain too broad, with few insights into employing GenAI in either upskilling or reskilling IT workforces in developing economies.

Despite these studies' valuable insights, significant gaps remain regarding the specific mechanisms through which GenAI can be effectively harnessed to address skills shortages within Colombia's technology industry. Current literature frequently provides general assessments of AI's potential rather than concrete, actionable insights. Moreover, insufficient research systematically addresses stakeholder perceptions, ethical considerations, and

specific strategies for scalable integration of GenAI into Colombia's distinctive socio-economic and regulatory contexts.

Therefore, this research aims to fill these critical gaps by conducting an in-depth analysis of GenAI's adoption in Colombia, especially regarding its application in IT workforce development, addressing unique local challenges and opportunities, evaluating stakeholder perspectives, and proposing practical frameworks for successful GenAI integration. In doing so, the study will provide helpful recommendations within a guideline specifically tailored for optimising workforce development strategies in Colombia's tech sector.

3 Research Methodology

3.1 Research Approach

In this study, an Inductive approach is applied to explore how GenAI shapes the upskilling and reskilling efforts in the technological industry. Instead of starting with a group of hypotheses, the study builds from the ground up, aiming to get insights from professionals who held different roles in IT Companies and professional education providers in Colombia. The objective is to discover firsthand experiences and perceptions about GenAI ´s role in workforce development.

Based on the principles of inductive reasoning, this approach enables key themes and patterns to emerge naturally from the collected data, thereby providing a stronger, more grounded, and detailed picture of reality. As an exploratory strategy, this approach is particularly well-suited for studying new technologies, such as GAI, alongside the perspectives of different actors. The goal is to develop insights and build theory from the bottom up, rather than verifying existing models. (Bryman, 2012)

Additionally, the researcher’s interpretation of the data was influenced by over eight years of professional experience in digital talent development, EdTech innovation, and technology-supported training systems. The researcher used their professional background to analyse and interpret participant responses, particularly when evaluating the complex aspects of the GenAI training program design and organisational implementation challenges.

The researcher's professional perspective aligns with interpretivist principles, as it strengthens the analysis by providing meaningful context for the data.

3.2 Research Design

The research design is qualitative and exploratory, grounded in a constructivist paradigm that assumes knowledge is co-constructed through social interaction and contextual

interpretation. This design is suitable given the novelty of GenAI and the complexity of its adoption in workforce development, particularly in emerging economies such as Colombia.

Semi-structured interviews were conducted with a diverse group of stakeholders directly involved in talent development processes, including technology professionals, educators, and leaders in learning and development (L&D). This design enabled flexibility to explore both shared patterns and context-specific insights across different institutional settings.

As Creswell(2013) explains, qualitative research seeks to understand the meanings individuals construct from their experiences, making it suitable for studies that aim to uncover emerging social and technological dynamics. Thematic analysis was conducted using inductive logic, allowing key insights to emerge organically from the data and be grouped into conceptual categories that directly respond to the research questions.

3.2.1 Data Collection

The instrument used to collect primary data for the research was semi-structured interviews, as they offer flexibility to adapt questions based on how the conversation naturally unfolds. This method is well-suited to the goals of my study, which seeks to explore the perspectives and experiences of professionals and educators regarding the use of Generative AI in upskilling and reskilling. It allows participants to reflect freely on their interactions with AI tools, the challenges they've faced, and the opportunities they've identified, offering richer insights into a complex and evolving topic.

According to Bryman(Bryman, 2012) and Galletta(Galletta & Cross, 2013) , this method provides a deep understanding of complex phenomena and individual perspectives, while allowing for structured inquiry and flexibility to explore emerging themes.

A total of 13 interviews were conducted virtually with tech professionals, HR and talent development professionals, and representatives from educational institutions in Colombia. This purposeful sampling aimed to capture diverse perspectives relevant to the research (Patton, 2002).

Roles	Number of Interviews
Tech Professionals	6
L&D/HR Professionals	4
Educational Institution Professionals	3
Total	13

Table 2. Interview Role Distribution

The research participants were selected for their professional positions, which aligned with the study's research goals. The three key stakeholder groups included:

- **Technology (Tech) Professionals:**
  Participants in technical positions at software companies and IT firms include developers, engineers, and data analysts. The participants provided direct insights into present-day skill requirements and their individual training experiences, explaining how generative AI tools impact the upskilling and reskilling processes.

- **L&D/HR Professionals:**
  The group comprises Human Resources managers, talent development specialists, and Learning and Development (L&D) professionals who lead the planning and execution of training programs. The organisation's AI-based workforce strategies, internal barriers to adoption, and future talent needs became clearer through their insights.

- **Educational Institution Professionals (Educators):**
  The participants from this group worked at higher education institutions or vocational training centres as faculty members, academic coordinators, and curriculum designers. The interviewees provided essential insights into the integration of AI technology in educational programs and institutional training versus labour market needs, as well as challenges in expanding AI-driven learning models.

The participants were selected through purposive sampling, based on their active involvement in GenAI-related training practices or strategic decision-making regarding digital talent development. Most were approached through professional networks or direct institutional outreach.

Interview durations ranged from 20 minutes to 1 hour and 44 minutes, with an average of approximately 55 minutes. All interviews were transcribed verbatim and anonymised when requested by participants, before analysis.

Organisational Context of Participants

The selection of participants reflects a deliberate effort to include voices from a range of institutions that are actively shaping Colombia's digital talent development ecosystem. As you can see in Table 3. The Educational Institution Professionals include representatives from three distinct organisations: one is a leading EdTech company that offers accessible, high-quality, and job-oriented IT training programs to young Colombians, operating outside the formal university system but playing a critical role in the insertion of an agile workforce. The other two participants are affiliated with two of Colombia's most prestigious private universities, known for their innovation-oriented programs in engineering and technology.

The Tech Professionals come from a diverse group of technology companies. Their roles range from software developers to technical team leads, providing insight into both the operational and strategic aspects of GenAI implementation. A significant number of them were affiliated with Company D, a large firm with multiple tech teams operating in different capacities.

The L&D / HR Professionals represent three major Latin American software companies. Two of these (Company A and Company B) are multinational firms with operations across several countries, offering regional and global perspectives on AI-driven training and organizational development. The third, Company C, is a well-established Colombian firm with strong national influence in the IT services sector, known for its commitment to upskilling initiatives and digital innovation.

This diversity of organisational contexts enabled the study to examine how different types of institutions, with varying missions, scales, and levels of digital maturity, approach GenAI adoption in training and talent development.

ID	Name	Organization	Type of Stakeholder	Duration Intereview (minutes)
1	Diana	Company A	L&D/HR Professional	0:28:22
2	Sandra	Ed-tech Company	Educational institution Professionals	0:20:00
3	Andres	Company A	Tech Professional	1:09:41
4	Maria	Company D	Tech Professional	0:43:37
5	Norha	ICESI University	Educational institution Professionals	0:41:01
6	Camilo	Company B	L&D/HR Professional	0:47:32
7	Jorge	Company D	Tech Professional	1:44:55
8	Isabella	Company C	L&D/HR Professional	1:02:02
9	Juan	Company C	Tech Professional	1:02:02
10	Gabriela	Company C	L&D/HR Professional	1:02:02
11	Jose	EAFIT University	Educational institution Professionals	1:03:00
12	Jessica	Company D	Tech Professional	0:45:36
13	Marcela	Company D	Tech Professional	0:53:36

Table 3. Interviewees characteristics

Interview Structure

It is essential that the interviewer asks benevolent questions and demonstrates an interest in the informant's story and experience. The interviews were conducted using an interview guide developed by the researcher (see Appendix C). This guide comprises approximately 16 questions, divided into seven blocks, as outlined in Table 3.

Interview Blocks	Number of Q
1. Introduction and Contextual Understanding	2
2. Learning Challenges and Current Upskilling/Reskilling Practices	3
3. GenAi adoption and Use	2
4. Challenges and Barriers	4
5. Opportunities, Best Practices and Recommendations	2
6. Ethical Concerns	1
7. Future Perspectives	1
Closing: Open Comments	1
Total	16

Table 4. Interview blocks

Each block represents a specific topic, with 2 to 4 questions covering each of the research questions and the research objective. The evidence of this is in Table 4. Also, Follow-up questions were asked throughout the interview to achieve depth in the informant's study (Ahrne & Svensson, 2015). This approach allows the researcher to revisit the topic with new questions and verify the informant's stories.

Interview Blocks	RQ1	RQ2	RQ3	Obj. 1 (Adoption)	Obj. 2 (Barriers)	Obj. 3 (Stakeholder views)	Obj. 4 (Roadmap)
1. Introduction and Contextual Understanding				✓		✓	
2. Learning Challenges and Current Upskilling/Reskilling Pra	✓	✓		✓	✓	✓	
3. GenAi adoption and Use	✓	✓	✓	✓	✓	✓	✓
4. Challenges and Barriers	✓	✓	✓		✓	✓	✓
5. Opportunities, Best Practices and Recommendations	✓		✓		✓	✓	✓
6. Ethical Concerns	✓					✓	✓
7. Future Perspectives			✓			✓	✓
Closing: Open Comments	✓	✓	✓	✓	✓	✓	✓

Table 5. Relations among Interview blocks with RQs and Objectives

3.3 Data Analysis

The collected interview data was analysed using Thematic Analysis, a flexible and widely used method for identifying, analysing, and reporting patterns within qualitative data. The approach applied was both inductive and deductive, allowing for the emergence of grounded insights while remaining connected to the study's research questions and interview framework.

The first part of the analysis began by creating an initial codebook, which was informed by the interview guide structure and its embedded themes. The initial codes originated from the specific areas of investigation that were explored during the interview process. The analysis process introduced an inductive coding phase, which allowed the researcher to identify new

codes that emerged from the data through recurring concepts, tensions, or insights that became evident.

Each code was subsequently grouped into broader categories (parent codes) to reflect overarching patterns and relationships better. Additionally, each code was systematically connected to the relevant research question. This alignment process enabled the researcher to perform a structured and strategic synthesis of findings between different cases and participant roles.

While the complete codebook can be found in the appendices (Appendix D), the table below (Table 6) presents a portion of the final codebook. It demonstrates how codes were organised into categories, which were in turn linked to the given research questions, RQ1, RQ2, RQ3, allowing for an organized interpretation of the data across conceptual layers.

Code	Parent Code	Related RQs
Absence of GenAI in Training	GenAI Adoption & Implementation Challenges in Training and Talent Development	RQ1
Organizational Readiness for GenAI in Training	GenAI Adoption & Implementation Challenges in Training and Talent Development	RQ1
Corporate Training Strategies	Learning & Training Approaches	RQ1, RQ3
Learning Strategy Orientation	Learning & Training Approaches	RQ1, RQ3
Learning Practices	Learning & Training Approaches	RQ1, RQ3
Training Delivery Constraints	Learning & Workforce Development Needs	RQ1, RQ3
Alignment with Local Context	Stakeholder Perceptions	RQ2
Equity and Inclusion Views	Stakeholder Perceptions	RQ2
Ethical Concerns	Stakeholder Perceptions	RQ2
Perceived Efficacy of GenAI	Stakeholder Perceptions	RQ2
Trust in AI for Training	Stakeholder Perceptions	RQ2
Vision for Strategic AI Integration	Stakeholder Perceptions	RQ2
Community-Based Innovation	Strategic Solutions for GenAI in Talent Development	RQ3
Partnership Opportunities	Strategic Solutions for GenAI in Talent Development	RQ3
Proposed Strategies	Strategic Solutions for GenAI in Talent Development	RQ3
Scalability and Sustainability Models	Strategic Solutions for GenAI in Talent Development	RQ3
Skills-to-Employment Alignment	Strategic Solutions for GenAI in Talent Development	RQ3
Successful Use Cases / Pilots	Strategic Solutions for GenAI in Talent Development	RQ3

Table 6. Fragment of the codebook

The next step of the coding process was the development of core themes, which emerged through repeated analysis and constant comparison of participant testimonies. These themes represent general conceptual categories that combine multiple codes to show standard patterns related to GenAI adoption, training strategies, stakeholder perceptions, and implementation practices.

The thematic analysis produced common findings which demonstrated both institutional and role-based similarities and differences. These findings are visually summarised in the diagram below (Figure 1), which presents a visual representation of

the coding categories that evolved into thematic areas, ultimately producing empirical findings. The following chapter (Chapter 4: Findings) presents detailed findings for each theme, accompanied by supporting quotes and interpretive analysis.

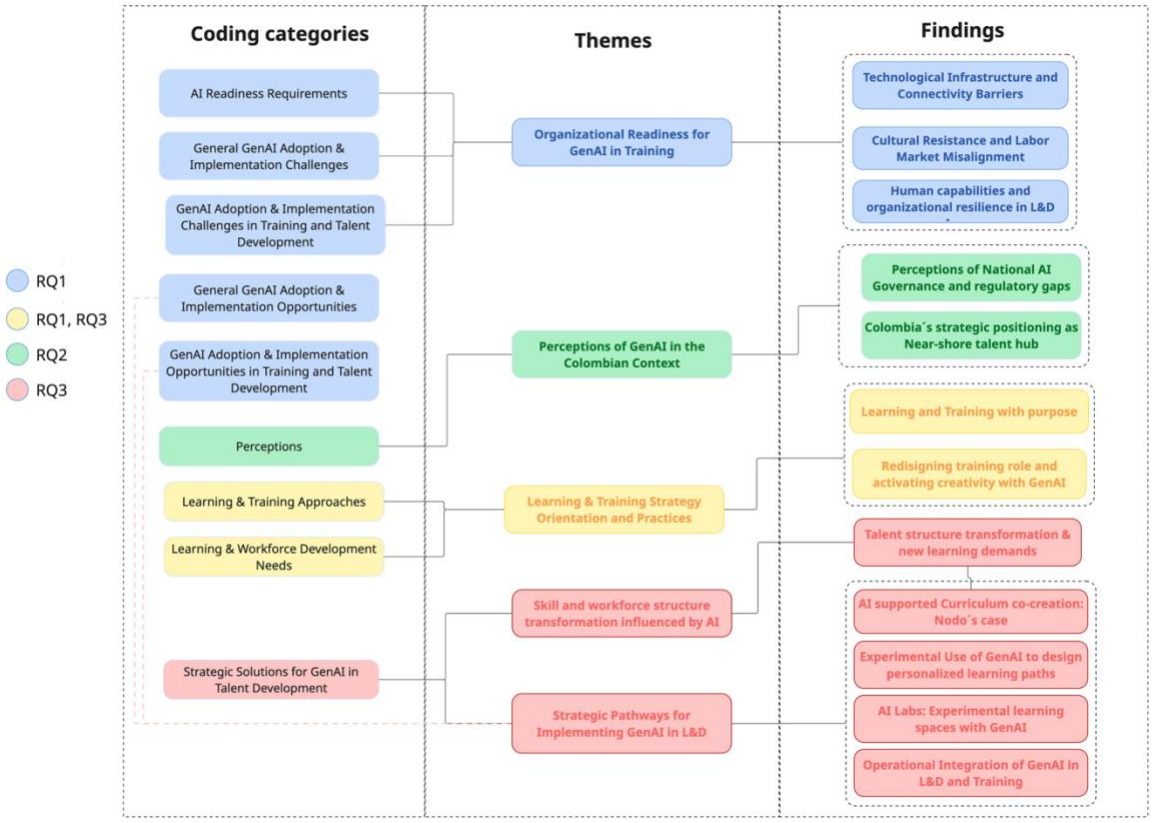


Figure 1. Thematic Analysis Journey

3.4 Ethical Considerations

Multiple ethical measures were considered to safeguard participant rights, welfare, and privacy in this study, while adhering to academic research standards and ethical guidelines.

- **Informed Consent:** The experimental study participants received detailed information about the research before each participation. The interviewees received comprehensive details about the research objectives, along with detailed descriptions of the procedures and boundaries, as well as information about possible risks and expected benefits, and a clear understanding of their role in the study. The information was presented in a clear and accessible manner. The participants received an informed consent letter (*Appendix C*), which they were required to read and sign to demonstrate their voluntary participation in the study.

- **Confidentiality and Anonymity:** Research participants received assurances that their individual information would be kept confidential, as requested. The researchers stored personal data independently from research findings on a secure local server. The research data will be stored for six months after study completion before being completely deleted. The research reports used pseudonyms instead of actual names to maintain participant anonymity.

- **Right to Withdraw:** Every participant received clear information about their right to leave the study at any time without facing negative consequences. A system was in place to handle participant withdrawals through procedures that minimised disruptions to research schedules. When participants end their study participation, their related data will be erased from the dataset.

4 Findings

4.1 Theme 1: Organisational Readiness for GenAI in Training

4.1.1 Technological infrastructure and connectivity barriers to GenAI adoption

One of the most persistent and foundational barriers to the adoption of Generative AI (GenAI) in workforce training in Colombia is limited access to technological infrastructure, particularly in underserved regions or among populations with fewer digital resources. At the same time, much of the discourse around AI integration assumes baseline access to connectivity and devices, specifically in underserved populations with limited digital resources. Most conversations about integrating AI into a system are based on the premise that there exists digital connectivity and access to devices. The realities shared by participants show that those assumptions do not apply to large segments of the population.

This concern was particularly evident among participants from the education sector. José (Educator, EAFIT, 2025) pointed out that, *"the challenge is access to technology."* He mentioned that many educational spaces do not provide reliable internet connections nor fundamental devices which prevents the implementation of GenAI-supported learning systems. He added: *"In many territories the challenge remains that there is no good connectivity, or no devices. Even when cellular is available, data is not always available."* This highlights the fragility of digital infrastructure as a prerequisite for AI-enabled training, particularly in areas beyond major urban centres.

The issue is not limited to access, but extends to infrastructural adaptation. To address this issue, José shared the case of Hacku, a startup using WhatsApp and microlearning capsules to deliver training in low-connectivity environments. This demonstrates that organizational preparedness for GenAI adoption goes beyond resource availability; it includes the ability to adapt the learning design to the context.

These concerns were echoed by Diana (L&D Professional, Company A, 2025), who mentioned: *"There are many people who don't have a digital education background... and that's needed to be able to use AI tools."* Her observation reflects how technological barriers are intertwined with digital literacy gaps, which further complicate the implementation of GenAI-based systems.

Jorge (Tech, 2025) emphasised the physical nature of this problem when he stated, "The construction of artificial intelligence requires more than discussions... machines and hardware are necessary." His perspective introduces an additional consideration to the discussion, as without suitable technical equipment, AI-enhanced training discussions may lose their operational value.

Together, these insights reveal that infrastructure barriers play a fundamental role in determining GenAI readiness. They demonstrate that the success of AI-supported workforce development strategies in Colombia depends on the availability of tools, alongside institutional capacity to bridge access gaps and develop digital competencies, as well as adapt technological capabilities, especially when operating in unequal contexts.

4.1.2 Skill recognition gaps in the labour market

Aside from the lack of infrastructure or technical capacity, a key challenge to GenAI adoption in training lies in the disconnect between how skills are developed and how they are recognised in the labour market. As José (Educator, EAFIT, 2025) emphasises, "we have not yet moved to the skill-first approach", highlighting the fact that most companies still use formal qualifications as the primary filter.

This resistance to recognising alternative learning paths limits the impact of GenAI in training. Even when these tools are efficient and provide tailored upskilling, their adoption is restricted due to employers' over-reliance on credentials.

While other participants did not explicitly state this, policy documents at the national level, like the *Roadmap for the Ethical and Sustainable Adoption of AI* (MinCiencias, 2024) and CONPES 4144(DNP, 2025) , emphasise the importance of aligning talent

development with the real needs of the digital economy. Without transforming hiring practices and skill recognition approaches, GenAI will have a limited impact.

4.1.3 Avoiding the hype: Constructing shared understanding in GenAI adoption

The interviews revealed a subtle yet essential challenge in GenAI adoption: fostering a collective knowledge of the technology that extends beyond hype or casual enthusiasm.

José (Educator, EAFIT, 2025) warned against jumping into innovation without reflection: "*The fashion was that everyone was going to the metaverse… no, come on, this is not about fashion, this is about understanding what's going on.*" This highlights the risk of treating GenAI as a passing trend rather than as a tool that requires deep understanding and context for practical application.

To address this, José cited *Magnolia*, an expert roundtable hosted in Medellín (Colombia) that prioritises collective learning and critical exchange as an example, "What I like the most about the gathering is that people of such a high level go there that I leave learning… one needs to be challenged" He argued these spaces are crucial for developing the shared concepts and structures necessary to design or realign educational programs involving GenAI.

Jorge (Tech, 2025) also raised this concern, noting that countless organisations invest in emerging technologies with no clear direction: "Many companies confuse fashion with trend, and end up investing in pilots that last only three months and are then forgotten." His remark serves to exemplify the problem of reactive experimentation, where pilots are executed without any systematic alignment with overarching strategic goals.

Diana (L&D, Company A, 2025) offered a similar perspective from the corporate learning space: "Before designing a course or a use case, we need a clear conceptual framework. No one wants to repeat mistakes of trying out technologies because they're trendy." For her team, moving forward with GenAI requires shared understanding and internal alignment, not just access to new tools.

These observations reveal a notable gap that is not technical, but rather cognitive and cultural. In the absence of structured frameworks that facilitate contemplation and collaborative learning, initiatives involving GenAI are likely to become either standalone projects or responses to market pressure, rather than integral components of a sustainable educational transformation strategy.

4.1.4 Human capabilities and organisational resilience in talent areas

The study revealed an ongoing issue, as teams responsible for training initiatives face substantial barriers in implementing Generative AI, despite its proven value for recruitment, learning path design, and content generation.

As Camilo (L&D, Company B, 2025) stated, HR professionals possess deep relational and human skills, but often maintain a distance from digital tools that could provide support in their work. "*They're great at the human side, but being so far from these tools, it's hard for them to know how to use them [...] there's a kind of apathy.*"

Diana (L&D, Company A, 2025) explained that organisations tend to view GenAI as exclusively belonging to IT, which produces mental resistance to broader usage. *"Some say, 'that's technical, it's not my thing' [...] they don't see how it could help them in their role."*

According to Gabriela (L&D, Company C, 2025), the communication gap results in employees showing minimal interest in the tools and failing to provide feedback, hindering progress through iterative refinement. *"They're not really into giving us feedback [...] and that makes the adoption process slower."*

The disconnect exists beyond human resources teams. Juan (Tech, Company C, 2025) emphasised that human oversight, alongside data integrity maintenance, is crucial for GenAI implementation. Without these, automation can quickly lead to poor or even harmful decisions.

A similar tension exists in the education sector. Norha (Educator, ICESI, 2025) noted that university professors with access to necessary infrastructure often face difficulties in understanding how AI works, while also struggling to consider its broader implications. *"AI needs to be understood from its epistemology. But many teachers don't even get how it works, and that limits their ability to use it in education."*

The organisational barriers receive recognition from policymakers at this stage. The *"AI Adoption Roadmap*(MinCiencias, 2024) encourages educational institutions to take the lead in professional reskilling and upskilling initiatives, transforming professionals into active changemakers within their organisations. The data on educator digital skill levels remains insufficient, according to CONPES 4144(DNP, 2025), which obstructs the creation of targeted strategies for adopting emerging technologies in teaching and learning.

All these points suggest that GenAI readiness is not just about infrastructure, vision or leadership. It also requires continuous efforts in both internal capacity development and cultural evolution, as well as cross-departmental collaboration, specifically targeting departments that tend to resist technological innovation.

Closing reflection

The findings discussed in this theme make it clear that being "ready" to adopt GenAI goes far beyond having the right tools or showing initial interest. The adoption of GenAI faces real barriers due, due to unequal access, resistance to change, a lack of shared language, and limited internal capabilities. Learning and Development teams demonstrate curiosity and enthusiasm, but they lack the readiness to take the lead. GenAI faces the risk of being introduced superficially because organisations don´t have unified strategies and direction. The willingness is there, but appropriate readiness is still a work in progress.

4.2 Theme 2: Perceptions of GenAI in the Colombian Context

4.2.1 Perceptions of national AI Governance and Regulatory Gaps.

Many interviewees expressed concern about the lack of a regulatory framework that guides the implementation of GenAI in Colombia, particularly for educational purposes. All professional groups stated that the Colombian government has published strategic documents, yet actual implementation and institutional structure development remain inadequate.

José (Educator, EAFIT, 2025) acknowledged the existence of the CONPES 4144 on AI as a step in the right direction, but emphasised its limitations: "*Colombia has just published the CONPES… I think it provides a clear direction on where we should go. However, we are still lacking*."

He also criticised government investments in infrastructure disconnected from scalable learning strategies: "*The money cannot be spent on creating a building and bringing in servers […] there is a lack of knowledge in the government."*

This lack of coordination and strategic clarity was mentioned by Camilo (L&D, Company B, 2025), who emphasised the need for a regulatory framework that understands how businesses execute AI implementation: "*It is a challenge because the regulatory issue is still very new, and I think the government does not even fully understand how this works. So, there is no clear guidance."*

From the Education sector perspective, Sandra (EdTech Org, 2025) expressed concern over the gap between technological advancement and public policy:
"*AI is moving much faster than public policy. And that generates fear, because we don't know what can and cannot be done."* Her comment highlights how unclear regulations lead to cautious and delayed adoption.

A similar concern was raised by Isabella (L&D, Company C, 2025), referring specifically to the education sector: "*There are no guidelines yet from the Ministry on how to implement this in education. Everyone is doing what they can, but there is a lack of guidance."*

These perceptions align with the limitations acknowledged in public policy documents. The AI Roadmap for Colombia(2024) highlights that the country lacks clear, up-to-date regulatory frameworks and specific institutions with the mandate to guide the adoption of AI across various sectors. Similarly, CONPES 4144(2025) highlights the absence of operational mechanisms to verify the ethical use of AI and the need for better articulation across ministries, especially in education.

In conclusion, these findings collectively suggest that while Colombia has articulated a strategic vision for AI, its regulatory and governance ecosystem remains fragmented and poorly connected to the training and education needs of the workforce. For GenAI to be implemented responsibly and effectively, stakeholders expect not only national guidelines but also practical, contextualised support from public institutions.

This issue will be revisited in the Discussion chapter, where we explore the broader policy implications and the need for coordinated action across sectors.

4.2.2 Colombia's strategic positioning as a near-shore talent hub

Interviews consistently highlighted that Colombia possesses key attributes that make it an attractive destination for global software companies, and by extension, for GenAI-driven workforce development initiatives. José (Educator, NODO, 2025) summarised this positioning as follows:

"International companies come because we're basically in the same time zone as them, and because Colombian developers work with great willingness... we have strong capabilities."

Comments from the private sector reinforced this perception. Jorge (Tech, 2025), a delivery manager at a tech consultancy, emphasised cost-efficiency as a decisive advantage:

"Foreign companies already understand the value of Colombian talent. Setting up an offshore operation with our professionals is a big plus."

From a human resources perspective, Diana (L&D, Company A, 2025) highlighted the benefit of continuous global coverage:

"We're a huge company with many employees worldwide, and from Colombia, we can support projects 24/7 thanks to our time zone and the number of available profiles."

These advantages are built around three main pillars: time zone alignment with North America, regional cultural affinity, and competitive cost structures. Together, they reduce friction in international collaboration, making Colombia an efficient partner for cooperation. As Jorge pointed out, lower labour costs combined with high-quality output make Colombian tech talent especially appealing for offshore operations.

Furthermore, the perception of commitment and technical capacity, reinforced by successful global project experience (as noted by Diana), contributes to positioning the country as a hub for advanced talent.

Taken together, these insights suggest that Colombia's strategic international positioning could be a key enabler for the adoption of GenAI in training, as it attracts investment, facilitates knowledge transfer, and expands opportunities for public-private collaboration.

4.2.3 Distrust and critical thinking regarding the use of GenAI in training

A key concern among participants from various sectors was the opacity of GenAI models, the potential biases they may contain, and the lack of effective regulation in the Colombian context. This distrust is not abstract; it manifests in the way actors understand and apply GenAI in their training and decision-making processes.

Camilo (L&D, Company B, 2025) was direct in pointing out that the country is not yet prepared to regulate this technology appropriately, and underlined the importance of not blindly trusting the responses of GenAI tools:

“It's not just going to be eating whole about what we're told, but it's also questioning: why that answer? Why is it white and not black?"

Julián (Tech, Company C, 2025) reinforced this concern from an organisational perspective, noting that even though his company has explored the use of GenAI in areas like talent selection, they’ve avoided automating critical decisions due to the lack of transparency and control over these models:

“We have avoided automating critical decisions [...] we don't want it to be seen as an oracle.”

From an ethical perspective, Gabriela (L&D, Company C, 2025) emphasised that the use of GenAI must always incorporate human criteria and frameworks for responsibility, especially when those decisions directly impact people. She thinks that it is an urgent matter for organisations to develop internal governance policies when external regulation is not yet in place.

Norha (Educator, ICESI, 2025) brought this reflection into the educational field. She insisted that the training of professionals using GenAI must be centred on ethical reasoning and critical thinking. For her, the danger is not just misusing the tool, but producing professionals who don’t learn to question what technology gives them:

“Students must be taught to doubt, to contrast, to not take for valid every automatically generated answer.”

Aleja (Tech, 2025) added that ethical frameworks should not be developed solely by legal or technical teams. Instead, she proposed including human and pedagogical perspectives from the beginning. She views HR teams as key players in shaping ethical usage policies for GenAI, particularly in training environments.

Finally, Andrés (Tech, Company A, 2025) highlighted that even tech professionals can be misled by answers that "sound right" but are ultimately wrong. This reinforces the need to validate and critically analyse the outputs generated by these tools:

“Sometimes even I get carried away because the answer sounds right, but it's wrong.”

In conclusion, these testimonies suggest that integrating GenAI into training and workforce development requires more than technical infrastructure or digital skills, even soft skills. It calls for building critical, ethical, and reflective capacities in both students and professionals, enabling them to interact with these technologies responsibly and autonomously. Furthermore, the lack of precise national regulation

intensifies the need for organisations to develop internal governance frameworks that promote the conscious, transparent, and fair use of GenAI.

From the interviews, I can infer that GenAI is not being received neutrally or passively. In Colombia, perceptions about this technology are influenced by a mix of opportunity and caution. While the country's global positioning and human capital offer prominent advantages, scepticism around regulation, ethics, and transparency is equally present. For GenAI to serve as a significant asset in training, it must be critically embraced, not simply as a tool, but rather as a system to be questioned, shaped, and controlled. These perceptions underscore the pressing need to foster trust, leverage latent capabilities, and establish a unified framework that integrates innovation with locally grounded contexts.

4.3 Theme 3: Learning & Training Strategy Orientation and Practices

4.3.1 Learning and training with purpose: Activating reflective motivation

One of the key barriers to implementing GenAI as a learning tool in tech organisations is the lack of motivation and purpose-driven actions among professionals. Camilo (L&D, Company B, 2025) emphasised that even when content, tools, and access are available, many people are simply not motivated to step out of their comfort zones to learn something new:

"One of the main challenges is not only with technology or lack of talent. It's more a matter of mobilising talent and interest [...] it takes a lot of thoughtful motivation to learn new things."

This lack of engagement does not manifest as active resistance, but rather as a form of inertia or apathy, where people choose to stay with what they know instead of making the effort required to learn new technologies, even when they recognise their value. It challenges the assumption that access to platforms or digital tools alone will lead to meaningful learning.

José (Educator, EAFIT, 2025) reinforced this view from an educational perspective. In his experience, many initiatives fail because the learning process lacks a clear and compelling purpose:

“The important thing is for people to connect with the purpose of what they are learning.”

When the learning journey lacks narrative or fails to align with individual interests or goals, it is perceived as a burden or just a formality. For GenAI to work as a meaningful training tool, the process must address not only what and how to learn, but also why.

From the non-formal educational perspective, Sandra (Educator, 2025) observed that students often disengage from GenAI-based courses when they don’t see how the content connects to their personal or professional paths:

“We try to use AI in our courses, but many students don't see the value in it if they don't understand how it helps them in their career or day-to-day life.”

Isabella (L&D, Company C, 2025) shared a similar reflection from the corporate side, highlighting how fragile learner engagement can be:

“There are people who start with a lot of enthusiasm, but if they don't understand what it's for or how to apply it in their role, they lose motivation very quickly.”

Even from a tech role, Jorge (Tech, 2025) shared that learning is often ineffective when there is no structure or guidance in place. For him, GenAI is a helpful tool when he uses it to organise his learning journey, but it is overwhelming when the content has no clear direction:

"Sometimes you don't even know where to start. If they show me a roadmap, I move forward. But if they throw all the content at me, I switch off."

These findings suggest that for GenAI to serve as a transformative tool, purposeful and constructive frameworks, alongside clear goals, must guide the learning process. Without proper frameworks in place, motivational scaffolding, or intentional design, the training mission risks reduction to mere automation rather than a process of meaningful growth and development.

4.3.2 Redesigning the training role and activating creativity with GenAI

The findings show that incorporating GenAI in training programs requires not only a transformation in content but also a shift in traditional teaching roles. Different types

of actors within the ecosystem agree that AI can serve as an intelligent assistant, facilitating the redistribution of tasks, designing new challenges for students, and activating more creative processes in both teaching and learning.

From the educator's perspective, José (Educator, EAFIT, 2025) described this potential clearly:

"You give the teacher time. You give the student with additional challenges and have an assistant. So, how can you implement creativity?"

From the perspective of a talent management professional (L&D-Talent Development), Diana reinforces this idea:

If we use AI only to speed up operational tasks, we lose the potential to develop critical thinking, creativity, and even leadership. We should free the trainer from mechanical tasks so they can focus on those skills that really transform."

Finally, from a tech professional's point of view, Jorge points out a complementary risk:

"AI has to help us design better, not do more of the same faster. If in the end it only serves to automate tasks without thinking differently, we are failing."

These voices converge on the need for GenAI not to be limited to improving operational efficiency, but to serve as a lever for redesigning the training process, promoting creativity, intentional pedagogical design and challenges adapted to more meaningful learning.

Closing reflection

The findings suggest that the training of GenAI is successful not merely because of technology or availability, but due to the planned learning processes, experiences, and guidance provided. When it is used appropriately, GenAI can support transformative change. It must be used purposefully, aligned with personal goals, and integrated into creative, well-structured learning environments. However, in the absence of design and apparent motivation, it would be no more than an efficient but empty.

4.4 Theme 4: Skill and Workforce Structure Transformation Influenced by GenAI

4.4.1 Talent Structure Transformation and New Learning Demands in Tech Teams

The interviews demonstrate a current transformation of roles, expectations, and skill demands within tech teams. The emergence of GenAI and other technological advancements is not simply introducing new tools; it represents a fundamental shift that transforms traditional technical job roles.

This aligns with what was discussed earlier in previous sections of this study: he tech industry faces a fundamental structural skill gap. The competencies once considered sufficient no longer meet the current demands. Organisations are now under pressure to upskill existing professionals and reskill others into entirely new functions, while redefining how talent is developed and deployed.

Jessica (Scrum Master and Project Leader, 2025) explained how her responsibilities have expanded beyond coordination and delivery, showing how tech leadership roles are also evolving to include data literacy and AI-related competencies:

"*I work as a Scrum Master and lead projects, but I have had to get into data analysis issues because now they ask us for more things. They expect you also to understand AI or how to use tools that we didn´t use before."*

Her experience demonstrates the growing need for professionals to develop hybrid roles that combine their core skills with AI competencies, data management expertise, and autonomous learning abilities.

From the HR side, Gabriela (HR, Company C, 2025) explained how recruitment priorities have also evolved:

"Today we are looking for people with hybrid profiles. It used to be enough that they knew how to program; now you need them to also understand the business and have analytical or even leadership skills."

This demonstrates how the industry is starting to adopt hybrid talent profiles that combine programming expertise with business acumen, as well as effective communication and strategic thinking skills, for leadership development and talent upskilling.

Andrés (Tech Professional, Company A, 2025) highlighted that learning itself has become a core requirement:

“Not only do the tools change, but the pace changes. Learning fast is a skill. Knowing how to search, knowing how to validate what you find. That's what I've had to improve the most.”

His observation suggests the emergence of essential meta-skills: the ability to learn, unlearn, validate information, and adapt in real-time. These are now indispensable in both upskilling and reskilling processes.

José (Educator, EAFIT, 2025) concluded by stating that these changes require organisations to establish new approaches for training:

“We have to train teams not only for the technical, but to be able to respond to change. It's no longer a list of courses, it's a different logic of training.”

Together, these voices reveal that GenAI drives fundamental changes in talent development strategies alongside technological transformations. The current industry demands that upskilling and reskilling efforts become mandatory, as roles change rapidly and learning must be continuous, integrated, and adaptable.

4.5 Theme 5: Strategic Pathways for Implementing GenAI in Talent Development

4.5.1 AI-Supported curriculum co-creation: The case of NODO at EAFIT University

A notable example of GenAI implementation in higher education comes from NODO, an innovation unit within EAFIT University. NODO utilises a hybrid, multi-phase process to integrate Generative AI into its curriculum design strategy, adapting to the changing needs of the tech workforce.

The process initiates with direct dialogue between technology and industry stakeholders. NODO conducts external advisory board meetings with private sector and tech industry stakeholders to determine present workforce needs, required skill sets and new occupational profiles before content generation begins. The instructional design process begins with these insights to ensure that educational objectives align with labour market demands.

Once the skill requirements are defined, the design team activates an in-house GenAI tool called "*Syllabus*," which generates a first draft of the training program. Based on entry and exit profiles, as well as clearly defined learning objectives, Syllabus produces a proposed structure that includes topics, practice hours, suggested learning materials, and overall content flow. This version is then reviewed and refined by the human instructional team, who ensure pedagogical soundness and alignment with institutional standards.

In the second phase, a separate GenAI model performs professional simulations in the target role to verify that the proposed curriculum enables learners to achieve corporate learning objectives (or occupational learning objectives). This validator system detects both missing content elements and conceptual gaps, as well as structural problems and unattainable expectations. The tool was initially introduced to counteract hallucinations and inaccuracies produced by earlier versions of the system.

Once the occupational alignment is confirmed, a third GenAI model takes over. This one focuses on the atomic generation of educational content, such as scripts, slides and detailed learning components, which the internal digital learning team can virtualise or instructors can use directly. This layer translates the validated structure into actionable instructional material ready for deployment.

This case illustrates a concrete and operationalised model of human-AI co-design for upskilling and reskilling, positioning GenAI not as a content creator in isolation, but as a dynamic assistant in an expert-led process. It demonstrates a strategic and scalable pathway for implementing GenAI in training programs, particularly in addressing complex workforce needs in Colombia's tech sector.

4.5.2 Experimental use of GenAI to design personalised career pathways

An emerging application of GenAI in talent development is its use in assisting the personalised design of career pathways. Camilo (L&D – Talent Development) shared a recent pilot within his company in which a customised GPT was explicitly developed for technology-related roles. The model allows employees to input a future development goal, such as becoming a senior developer within three years, and receive training recommendations aligned with that target:

"*A GPT created for roles in technology that could say, I don't know...if I want to be 3 years from now a senior developer, what do I need to learn, and it would give me all the recommendations*." (Camilo, L&D, 2025)

While still in an experimental phase, this use case demonstrates the potential of GenAI as a formative and professional planning assistant, capable of automating recommendations and structuring personalised upskilling or reskilling trajectories.

This vision is aligned with Diana´s (L&D, 2025) thoughts, who accentuated the need for building ecosystems that use data in designing adaptive training systems for specific roles' skill gaps, rather than merely using generic courses on training platforms:

"*Not only uploading courses to the platform, but understanding from the role what kind of competencies are needed and using AI to make that fit.*" (Diana, L&D, 2025)

From a tech professional's perspective, Jorge reinforced the usefulness of this type of solution:

"To have something that tells you: look, this is the roadmap for this role, it's amazing." (Jorge, Tech, 2025)

Finally, José (Educator, 2025) framed the discussion from a pedagogical angle, arguing that learning environments should be designed to empower learners to set their own objectives and create their journeys, thereby reinforcing the idea of GenAI as an enabler of learner autonomy and intentional learning design.

Together, these perspectives highlight a strategic opportunity: using GenAI not only to optimise training delivery but also to create individualised, role-based development plans in both corporate and educational settings.

4.5.3 AI Labs: experimental learning spaces with GenAI

In addition to formal training programs, some organisations have begun experimenting with GenAI through internal innovation labs, which serve as tools where employees can ideate and develop proof-of-concept (POC) projects. Camilo (L&D, 2025) discussed a research and AI lab initiative at his company, which seeks to foster applied experimentation on relevant business problems:

"*Because they are small in-house projects, where people have the luxury of learning"* (Camilo, L&D, 2025)

This method enables hands-on learning and problem-solving in a low-risk environment, while also serving as a strategic opportunity for innovation. In some instances, these experiments have transformed into marketable services.

"*More formal business has come out of this, where the POC closes, acquires a customer and so on."* (Camilo, L&D, 2025)

Even though these labs have significant potential, they face several challenges. In Camilo's case, the initiative was taken by the operations area instead of the talent development team, causing issues related to ownership, alignment, and cross-functional teamwork:

"*It's been more created and developed by the operations side [...] it's been quite an issue because it's almost like opening up a whole new area of responsibilities."* (Camilo, L&D, 2025)

José (Educator, 2025) highlights the potential of AI labs to facilitate learning, stating that authentic learning unlocks meaningful outcomes when participants are self-directed. Learners set goals, map out challenges, and pursue the desired endpoint.

"Sometimes we set up environments where the student sets a goal: I want to do this. And from there the path is designed." (José, Educator, 2025)

Similarly, Diana (L&D, 2025) also noted the value of secure yet challenging contexts for learning, where individuals can explore GenAI tools in connection with real-world challenges:

"*We want people to be able to explore new tools in a safe environment, but connected to real business problems." (*Diana, L&D, 2025)

Taken together, these experiences suggest that AI labs, when well-integrated into talent strategies, can serve as powerful sandboxes for learning and innovation. These lab systems provide professionals with the opportunity to safely experiment, fail, glean insights, and apply them in the real world, thereby catalysing technology adoption and competency development throughout the organisation.

4.5.4 Operational integration of GenAI in talent development and training.

Although there are several experiences of GenAI use in training and talent development, the most relevant finding in this area is not how advanced these practices are, but rather how limited and uneven their implementation remains within the Colombian context. The examined cases indicate that corporate learning environments exhibit low maturity in terms of GenAI adoption. Most organisations use these tools on a limited basis without standardisation or complete implementation.

These tools primarily emerge through independent pilots, initial trials, or internal programs that operate disconnected from core L&D strategies.

Camilo (L&D, 2025) shared an organised example that demonstrates GenAI application for content creation, along with survey processing and platform-based learning path development on Udemy.

"We have used it for learning content generation [...] for survey analysis [...] also to capture people's sentiment against X and Z [...] we use it for the creation of Learning Path on Udemy" (Camilo, L&D, 2025)

The described applications represent promising uses, but they are often experimental or early-stage implementations, driven by individual teams rather than part of unified learning policies.

The case of Company C demonstrates this general pattern in the industry. The tech company has not started using GenAI for training purposes despite its position in the field. The company is currently creating an internal AI laboratory, which will eventually enable future learning applications:

"*We are not yet using GenAI as a training tool. We are in the process of setting up the lab, and the idea is that from there we can transfer what we have learned to the training teams."* (Gabriela, L&D, 2025)

Even among those already experimenting with GenAI, the uses remain fragmented. As Diana noted, her team is still working on mapping competencies by role to adapt learning content better, but the integration is ongoing and incomplete:

"*We want to understand from the role what kind of competencies are needed and use AI to match that*." (Diana, L&D, 2025)

From the learner's perspective, Jorge emphasised how the presence or absence of structure in GenAI-powered training tools affects engagement:

"*If they show me a roadmap, I move forward. But if they throw all the content at me, I disconnect."* (Jorge, Tech, 2025)

In academic contexts, the use of GenAI tools, such as Syllabus at Nodo, demonstrates a more systematic implementation. However, even these cases often remain at the pilot stage or depend on internal champions rather than institutional policy:

"*We use a model that generates an initial draft of the training content based on the desired entry profiles and competencies.*" (José, Educator, 2025)

These cases reveal that in Colombia, GenAI is indeed entering the operational layers of L&D, but not yet as a consolidated, strategic layer. The potential is there, but its realisation will depend on wider coordination, policy alignment, and investment in both infrastructure and human capability.

5 Analysis and Discussion

5.1 From Readiness to Resistance: Structural Barriers to GenAI Adoption

While enthusiasm for the potential of Generative AI (GenAI) to revolutionise workforce development is growing, our findings reveal a disconnection between organisational readiness and existing structural obstacles that hinder its implementation. The Colombian ecosystem faces multiple concrete obstacles that extend beyond technological restrictions, despite official declarations of interest. The barriers to adoption consist of structural gaps in infrastructure, as well as organisational resistance and regulatory uncertainty, and, most importantly, the absence of pedagogical integration in training institutions and corporate learning systems.

Institutional fragmentation and innovation bottlenecks

One of the most recurring issues highlighted by both corporate and educational actors is the lack of intersectoral coordination and strategic clarity. Although Colombia has established national frameworks and public policies, such as CONPES 4144 (and the *Roadmap for the Ethical and Sustainable Adoption of Artificial Intelligence in Colombia*(Ministerio De Ciencia, Tecnología E Innovación, 2024), interviewees expressed that these policies are still far from being operationalised into actionable roadmaps at the organisational level. According to José (L&D, 2025), the pain of business has shifted, and organisations now require higher standards, but their policies do not always align with these needs.

The successful implementation of GenAI in adult education requires that technological possibilities align with institutional strategic plans(Storey & Wagner, 2024). GenAI tools

in Colombia exist independently from institutional training and innovation plans because many actors deploy them without strategic alignment.

Capacity constraints and the misalignment of internal priorities

Even in tech-forward companies, internal capacity remains a major constraint. Diana (L&D, 2025), an HR Project Manager, stated that operational priorities tend to overshadow internal innovation processes which have prevented meaningful integration of GenAI in talent training systems: "*Most efforts are focused on improving customer services, not internal processes... this takes away space to integrate tools such as GenAI in training.*". This is a sign of resource saturation and misaligned incentives, which Sidhu et al.(2024) refer to as *"institutional fatigue in contexts of accelerated technological change"*, rather than ideological resistance.

The patterns observed in universities are similar to those seen in other organisations. Norha, a dean in STEM disciplines at a major institution, states that faculty members often lack the time and support to redesign their curricula around GenAI: "That's a challenge, it's like the main challenge because you have to keep operating, but you have to transform everything and integrate that" (Norha, Educator, 2025). The results indicate that readiness depends on more than infrastructure and attitude, because it requires both organisational learning and governance, as well as sufficient bandwidth. This aligns with the concept of structural inertia described by Hannan and Freeman (1984), which suggests that organisations often resist change due to established routines and structures. Therefore, apparent innovation resistance stems from institutional bandwidth limitations rather than cultural unwillingness.

Organisations must have governance models which provide resources, time, and leadership capacity to support adoption and readiness to become meaningful. Studies on AI deployment in corporate and educational settings in Latin America confirm that strategic alignment is essential for transforming initiatives beyond exploratory and pilot phases(OECD/CAF, 2022)

Bridging the Skills Gap for GenAI Readiness

One of the most overlooked aspects of GenAI readiness is the growing disparity between current training practices and the rapidly evolving demands of the tech workforce. As shown in our findings, professionals are no longer expected to master static technical skills; instead, they must continually adapt to new tools, roles, and hybrid responsibilities. This transformation, partly driven by the pace of innovation enabled by GenAI itself, has intensified the skills gap across the industry.

Currently, GenAI is accelerating the shift in job descriptions and the demand for talent and skills, yet it ironically maintains the capacity to enable organisations to bridge that gap. If employed strategically, GenAI can aid in developing adaptive training frameworks that address evolving prerequisites more effectively, thereby making upskilling and reskilling initiatives more effective and scalable. Failing to recognise the profound change in skill distribution will lead to misalignment of training efforts, and worse, misaligned efforts will result in disconnected implementation of GenAI technologies.

As Tenger and Taeymans(2024) argue, bridging the skill gap requires more than technical integration; it demands a pedagogical and organisational redesign aligned with the shifting architectures of talent and the volatile demands of the digital economy.

Inequality in digital and conceptual readiness

Another barrier identified is the uneven digital and conceptual readiness across regions and roles. Diana pointed out that many employees struggle not with the use of GenAI itself, but with basic digital skills: “There are many people who don't have a digital education background... and that is needed to be able to use AI tools” (Diana, L&D, 2025). This digital gap is compounded by conceptual confusion, as Norha explained: “Many still confuse traditional AI with what GenAI can do today.” (Norha, Educator, 2025).

Storey and Wagner(2024) support these findings through their argument that implementing AI requires proper education about its theoretical and practical aspects, as otherwise, educators and HR professionals will either misuse or underutilise AI tools. The adoption process remains superficial because it stops at task automation but fails to unlock strategic innovation and deeper learning potential.

Human capital paradoxes: expertise gaps and resistance in HR

Perhaps one of the most revealing tensions lies in the paradox of human capital management: those who guide workforce evolution usually do not possess the competencies needed to use new technologies. According to Camilo (L&D professional, 2025), “They are very humanistic teams... it is difficult for them to know how to use these tools and how far they can go.”. This aligns with the research by Pavitra and Agnihotr(2023), which supports the idea that learning and development teams typically lack sufficient technical knowledge to effectively transform AI tools into valuable educational practices.

This paradox demonstrates how institutions responsible for AI integration lack the necessary capabilities to lead the process because they have not invested enough in their learning development and role evolution.

The GenAI landscape in Colombia reveals an essential contradiction, as organisations claim to innovate yet face internal barriers, combined with restricted capabilities and unresolved conceptual issues. Institutions need strategic clarity, technical skills, and pedagogical imagination to demonstrate meaningful action in readiness assessment. GenAI will persist in pilot projects, but remain disconnected from the more profound transformation of learning and workforce systems that it promises to enable.

These findings align with those of Sidhu et al.(2024) in other emerging economies, which suggest that the principal obstacles to AI adoption are not technological but organisational and pedagogical. Without strategic direction, internal alignment, and institutional preparedness, most GenAI endeavours are small in scale and disconnected from the broader transformation intentions.

This challenge is not only technical, operational or pedagogical, but it is also political. As Coeckelbergh(2022) explains, many decisions regarding the use of AI are made without public discussion or reflection. They are presented as purely technical steps, when in fact they carry significant ethical and educational consequences. In Colombia, the push to adopt GenAI often happens without asking basic questions: Why are we using it? Who decides how it should be used? And what values should guide its integration? When organisations treat GenAI as something that must be adopted quickly, without involving the people affected by it, they miss the chance to shape its role in a more thoughtful and transformative way.

5.2 The Role of Purpose, Engagement and Active Learning

The transformation of talent development through Generative AI (GenAI) necessitates its integration to support purpose-driven learning, active engagement, and thoughtful pedagogy, rather than replacing them. The interviews conducted in corporate and educational environments revealed that GenAI's value stems from its integration into learning spaces that promote agency, critical reflection, and practical problem-solving skills.

Purpose as a driver of motivation and deeper learning

Multiple interviewees stressed that purpose serves as the essential force which drives employees to maintain their commitment. In corporate settings, lack of adoption was not attributed to inadequate infrastructure, but to employees' disconnection from the

value of the training. As Camilo (L&D, 2025) stated, “The spaces are provided, the activities are carried out… but there is not really that much adoption. There is no real accompaniment, so this translates into learning.” Similarly, Jessica (Tech, 2025) noted that before learning any new skill, she needs to understand its purpose: “I need to be clear about what I want to learn something for…”.

The connection between learning and personal meaning through critical reflection aligns with Mezirow’s(1997) theory of transformative learning. The lack of purposeful connection makes GenAI-based training likely to become another efficient yet superficial intervention, rather than an effective one.

GenAI achieves its most significant impact when it adjusts to learner motivations while being integrated into environments that support meaningful development at both personal and professional levels. The ability to connect learning content to a meaningful purpose is essential for engagement in professional learning environments(BERSIN, 2018). As Storey and Wagner(2024) emphasise, this connection becomes even more crucial when aiming to foster autonomy and deep learning through GenAI-enabled systems.

The absence of a defined purpose represents both a teaching problem and organisational misalignment. The findings demonstrate that numerous organisations implement GenAI tools without developing a unified talent development strategy(Pavitra & Agnihotri, 2023; Sidhu et al., 2024). This disconnect leads to superficial adoption and undermines the long-term effectiveness of upskilling and reskilling training programs.

Active learning and the danger of passive automation

One of the most potent insights emerged from the Nodo initiative at Universidad EAFIT, where the entire training model is built around constructivist, challenge-based learning. José (Educator, 2025) stated that our organisation operates as a self-made system because people understand concepts through creation rather than passive instruction. This GenAI-supported, challenge-based learning model aligns closely with Kolb’s (1984) experiential learning theory, which emphasises learning through action, experimentation, and reflection.

This also aligns with Luckin’s (2018) concept of “intelligence augmentation,” where GenAI is not intended to replace human cognition but to extend and augment it. When embedded in well-designed learning ecosystems, GenAI can support learners in taking risks, generating ideas, iterating through feedback, and developing autonomy.

However, many companies introduce GenAI tools without proper pedagogical planning, which may promote “technocentric dependency”, a situation where tools

end up leading the process instead of supporting it. (Storey & Wagner, 2024). Such approaches may lead to the erosion of critical thinking, automation replaces curiosity, and learners are encouraged to follow rather than question. Without opportunities for experimentation, failure, and guided reflection, learning may become more efficient, but ultimately shallow, and most importantly, less effective.

This risk is not just practical but also more profound and philosophical. Heidegger (1977) reminds us that adopting technologies without questioning their role shapes our perception of the world. In education, this means that if GenAI is used only to speed up or simplify learning, we might start treating learning itself as just another output, something to measure, automate, and optimise. The real danger is not in the tool, but in letting it quietly redefine what it means to learn. Moreover, if we are not intentional, we may lose what GenAI could help us strengthen: reflection, autonomy, and the ability to think critically.

Designing for engagement, not just efficiency

To counter this, training programs should be explicitly designed to cultivate engagement, reflection and autonomy. Engagement, as argued by Luckin(2018) is not just behavioral compliance but a cognitive-emotional process that arises when learners are both challenged and supported. This is especially true in the context of digital talent development, where learners are required to develop technical skills alongside adaptability, ethical reasoning, and innovation capacities.

Several strategies can support this goal. The combination of GenAI with project-based learning, facilitated by AI assistance, enables learners to develop technical skills while practising creativity and autonomy(Storey & Wagner, 2024). Likewise, the use of GenAI agents in debate simulations helps students develop critical thinking and communication skills, and GenAI-generated feedback in reflection activities supports metacognition and improvement.

These approaches, applied with pedagogical intention, use GenAI not to replace effort but to empower learners in constructing knowledge and shaping their own developmental paths.

The transformative potential of GenAI, applied to upskilling, emerges from its strategic placement within pedagogical systems that emphasise purpose, engagement, and action. The tech industry of Colombia, along with its education sector, needs to avoid using AI as a simple replacement solution. Instead, GenAI should be integrated into systems that enhance human agency while providing adaptable learning routes and developing a professional environment that promotes curiosity, reflection, and purpose. The technology will speed up processes, but it will not lead to workforce transformation unless proper alignment exists.

5.3 GenAI and redesign of learning ecosystems

The rise of Generative AI(GenAI) is not only an instructional innovation; it presents a structural inflexion point for how learning is conceptualised, delivered, and managed. In both academic and corporate domains, GenAI prompts a reorganisation of curriculum architectures, institutional roles, and knowledge flows, necessitating a critical examination of whether existing learning ecosystems are fit for purpose in a rapidly evolving technological landscape.

From static curricula to adaptive learning paths

The interview data demonstrate that organisations are moving toward flexible and customised learning approaches. At Universidad EAFIT's Nodo José explained how their educational programs transitioned from traditional fixed curricula to collaborative student-educator-AI pathway development for personalised learning paths: "We design the route with the student according to what he/she needs to learn, and then we see how GenAI can accompany, suggest, correct, etc., and we can help the student to learn." (José, Educator, 2025).

GenAI technology supports the evolution from traditional fixed-course delivery to flexible, modular learning environments that adapt to individual student needs(Storey & Wagner, 2024) . These learning spaces emphasise both flexibility and immediate response capabilities to meet current learning requirements. The system enables skill development at the exact time it becomes necessary, while maintaining alignment with labour market requirements.

However, as Diana warned, many corporate learning structures remain outdated: "*There are loose initiatives… there is no structure to connect training with what is happening on a day-to-day basis*" (Diana, L&D, 2025). GenAI will become a superficial solution for rigid processes that fail to address complex upskilling requirements unless organisations completely transform their training systems.

Rethinking institutional roles and ecosystems of practice

The adoption of AI-assisted learning systems creates new distribution patterns for educational and professional development responsibilities. Jessica (Tech, 2025) explained her learning process by stating that she searches for courses independently, while AI provides direction, so she no longer needs instructions from the company about what to study. The GenAI-powered self-directed learning approach challenges traditional centralised training systems, suggesting the need for hybrid models that combine institutional guidance with the autonomy of the learner.

Nevertheless, the full potential of GenAI is easily undermined when institutions fail to integrate it structurally (Guettala et al., 2024). Implementing it in a fragmented manner not only limits outcomes but also reinforces inequality, as only well-resourced teams can effectively navigate and adapt to the new tools.

The organisational layer: beyond training

The Buelvas Parra et al.(2024) analysis of AI in Colombian organisations provides a critical connection. It reveals that many companies approach AI adoption without a parallel redesign of internal learning structures or workforce development strategies. The outcome results in operational efficiency improvements, yet capacity development and innovation processes remain stagnant. The authors demonstrate. The authors highlight how this process creates bigger employability gaps because workers need modern skills yet educational institutions do not update their curricula to match technological shifts.

This reinforces a key insight: redesigning learning ecosystems needs to transcend traditional educational institutions. It must extend to organisational cultures, HR systems, and public-private collaborations. As Pavitra and Agnihotri(2023) suggest, L&D teams need to transform into strategic learning ecosystem orchestrators, who will utilise GenAI not only for content creation but also for learning analytics, workforce planning, and personalised recommendations.

However, this transformation is not automatic. As highlighted in earlier sections, many HR and L&D teams encounter a paradox: they need to drive GenAI innovation yet they lack both digital fluency and pedagogical expertise (Camilo, L&D, 2025). If these teams are to guide the redesign of learning across entire organisations, and specifically address the upskilling needs of tech professionals, they must be the first to recognise, understand, and engage with the capabilities and limitations of GenAI. Rather than passive implementers, they must become active learners and advocates, building the internal literacy needed to anchor change from within.

So, GenAI offers the tools to reinvent how learning happens, but the fundamental transformation lies in how ecosystems are restructured to support it. In Colombia, this means the need to build systems that are modular, participatory, and integrated across educational and organisational domains. Without this redesign, GenAI may improve access and personalisation, but it will not close structural gaps in workforce development. Ultimately, achieving actual impact, at least organizationally, hinges on our institutional imagination and governance to reconfigure the system itself.

5.4 GenAI as a strategic tool for personalisation and skill development

Generative AI (GenAI) is emerging as a strategic infrastructure for Learning and Development in the tech sector. GenAI brings more value than content creation or automation because it enables customised learning routes and upskilling acceleration and practical training that matches individual objectives and company requirements. This section explores how GenAI can be, or is being, a catalyst for customised skill development in the Colombian context.

Strategic personalisation for dynamic talent development

The adoption of GenAI enables organisations and educational institutions to move beyond traditional one-size-fits-all training models by implementing this technology, which enables adaptive and modular systems. Advanced academic programs, such as those developed in innovative centres like Nodo, use GenAI to support learners in co-creating their development paths. These trajectories are aligned not only with disciplinary content, but with new roles that exist in the tech sector.

In corporate settings, meanwhile, this potential remains underutilised. Although some HR managers understand GenAI's diagnostic and recommendatory capabilities, they often limit their use to separate pilot projects that are disconnected from broader talent strategies. This underscores the need for integrating pedagogical and operational systems, where GenAI supports development aligned with both individual learning needs and real-time market demands(Storey & Wagner, 2024; Tenger & Taeymans, 2024).

Strategically, GenAI enables companies to close the gap between performance expectations with skill readiness, so addressing a major concern in contemporary digital environments.

The evolving role of L&D: from facilitators to capability architects

Learning and development (L&D) teams must go beyond content curation if they want to realise their potential. According to Bersin (2018), contemporary corporate learning strategies aim to sustain ongoing development through systems that integrate learning experiences, relevant data, and contextual performance support. For GenAI to be more than an accessory, L&D must become architects of personalised learning infrastructures, designing systems where GenAI supports, rather than substitutes, reflection, autonomy, and growth.

This shift doesn´t come without friction. Numerous HR teams understand their lack of readiness to drive organisational changes because they lack experience with GenAI tools and their educational effects. Here, the paradox is evident, and I have exposed it

in earlier sections: teams that need to provide upskilling opportunities also require it for themselves first. Addressing this knowledge deficit requires significant economic investments in the creation of GenAI literacy programs for HR and educational leadership.

GenAI integrated into learning experiences enables personalised student progress through tailored pathways that maintain both learner control over their work and awareness of their thinking processes.

Internal labs and situated experimentation

Some of the leading tech companies interviewed have started to develop internal labs and sandbox environments to initiate GenAI experimentation. These spaces serve as low-risk platforms where professionals can test GenAI tools in authentic contexts. However, several stakeholders noted a common barrier: formal training systems and performance evaluation frameworks do not align with these environments, resulting in a restricted strategic impact.

In order to unlock their potential, these labs must evolve into organised environments for applied learning, where GenAI is used not only for exploration but also to develop and assess relevant workplace competencies(Guettala et al., 2024; Kolb, 1984). This aligns with the *Roadmap for the Ethical and Sustainable Adoption of AI in Colombia* (2024) which proposes using AI sandboxes as a potential component of national innovation and skills development strategy.

Strategic value and contextual relevance in Colombia

The benefits of adopting GenAI as a strategic learning orchestrator are multiple. By automating skill diagnostics, speeding up content creation, and facilitating individualised learning experiences, it provides organisations with affordable, scalable, and data-driven alternatives to traditional training.(Tenger & Taeymans, 2024) These systems also contribute to talent retention by offering professionals clear growth pathways aligned with internal mobility and long-term career progression.

The effects of such implementation methods are more pronounced in Colombia. The existing digital talent development challenges stem from insufficient coaching opportunities, inadequate career guidance, and limited access to continuous training services, which primarily benefit elite groups. GenAI presents a possibility to democratise learning because it can deliver customised assistance to many people while bypassing traditional authorities to provide educational opportunities that were previously unavailable.

Although the value of GenAI for learning path customisation and enhancing professional growth is considerable, this value becomes even more urgent when viewed through the lens of Colombia's growing skills gap in the tech sector. According

to the Latin American Artificial Intelligence Index, Colombia scored only 25 out of 100 points in the workforce skills component, one of the lowest ratings in the region(CENIA, 2023)

As our findings indicate, the expectations on professionals are changing rapidly, requiring hybrid profiles, agile learning capacities, and constant adaptation to new tools and roles. GenAI has the potential not only to enhance training processes but also to serve as a structural solution through agile upskilling and reskilling, addressing the acceleration of the tech divide. This shifts the discussion from isolated learning improvements to a broader strategic agenda for talent development in emerging economies.

Aligned with public policy priorities and frameworks, such as the *Roadmap for the Ethical and Sustainable Adoption of AI(2024)* the strategic use of GenAI in workforce training presents an opportunity for collaborative innovation among the private sector, educational institutions, and government. Its potential to enhance personalisation and streamline training aligns with long-term goals of improving employability and digital inclusion.

Implementation strategies to activate GenAI potential

Organisations can use several essential methods to transition their experimental phase into strategic deployment of GenAI in upskilling and reskilling:

- **AI-powered diagnostics** to identify individual and team skill gaps, enabling them to create individualised learning plans based on actual development requirements.
- **Personalised learning journeys**, where GenAI supports the design of role-based pathways that match organisational goals, are especially useful in fast-moving areas like cloud infrastructure, data science or software development.
- **Human-in-the-loop models**, combining GenAI-generated feedback with coaching and peer-to-peer interaction, will guarantee that learning remains reflexive and context-sensitive.
- **Cross-functional collaboration** and strategic learning development require HR, tech leaders, and management roles to work together in the co-design of scalable strategies that align with business goals.
- **Public–private partnerships** will help to create interoperable AI-powered training systems that function together.

These strategies provide a solid foundation for integrating GenAI into workforce development processes. Their practical application will be further developed in the final adoption guidelines resulting from this research.

To sum up, if it is used strategically, GenAI is more than a tool; it is a platform for transformation. The primary strength of Genai lies in its ability to enable learners to

handle complex situations autonomously while maintaining relevance and achieving their objectives. In the case of Colombia, the challenge is not simply adopting GenAI, but rather adopting GenAI unified with integrated learning architectures that address systemic inequalities and changes in skill demand. The path toward the successful integration of GenAI requires institutions to envision new solutions while working across sectors and maintaining a commitment to inclusive practices, so that no one is left behind in the future of work.

5.5 Ethics, trust and critical thinking in training powered by GenAI

As workforce development is increasingly driven by the integration of Artificial Intelligence and GenAI, ethical challenges arise that are both complex and poorly explored, yet one that is affecting not just how systems get deployed, but how they condition the mindset and sense of agency of those who employ them. In professional training, specifically in the technology domain, where GenAI is being introduced to speed up processes and automate training, questions about trust, responsibility, and critical thinking become central.

Whereas previous sections of this study have examined how GenAI may be used for personalisation and skill-building on a strategic level, this part of the discussion is concerned with the essential tension of risks of uncritical reliance, disinformation, and opacity, and the immediate necessity of creating ethical awareness and critical thinking in users of such technology in professional contexts.

Ethical considerations as a new but unstructured problem

Throughout interviews, ethical tensions appeared not in the form of direct rejections, but rather as open-ended, unresolved questions. HR professionals were unsure what would constitute an ethical use of GenAI in the context of learning. Many acknowledged applying GenAI to automate training processes or content production, but did not consider when or how to reveal its use. Others mentioned a lack of mutual ethical standards: "*There is a vacuum, one does not know if it is right or wrong to do it, and no one is teaching that*", reported one interviewee.

Educators were also concerned about this. They observed that students were increasingly accepting GenAI responses without bothering to verify their accuracy and having the system decide for them. As one stakeholder put it, "*people stop thinking because the tool does it for them."* This shows a deeper problem: how do we ensure that GenAI augments human judgment rather than replacing it?

This is where Alvarado's(2023) contribution comes in. He argues that the only kind of trust that should be allocated to AI technologies, if any at all, is epistemic trust, trust based strictly on the system's capacity to provide information or convey knowledge In learning contexts, this requires that the learner knows how to use GenAI and when and

why one should distrust it. Ethical education is therefore not just risk mitigation; it is about creating a mindset in the learner that can evaluate the credibility and utility of AI-generated material responsibly, cultivating the habit of doubt and reflection.

The pedagogical imperative for critical and political awareness

Developing this kind of critical awareness requires more than tool tutorials. These concerns pertain to the citizen and professional we are cultivating. Therefore, it necessitates rethinking the pedagogical role of GenAI itself. Mezirow(1997) suggests that transformative learning requires something more than just new knowledge; it involves challenging assumptions, critical reflection, and informed action. However, this sort of agency is hard to cultivate if GenAI is employed in a manner that emphasises automation rather than questioning.

According to Luckin (2018), AI is intended to be a tool for intelligence augmentation, rather than a means of avoiding cognitive effort. When students begin to rely on GenAI as the primary source of knowledge, with limited understanding of its internal processes or how it may encode bias, the result is passivity, not empowerment.

This was echoed by Heidegger(1977) a long time ago. He cautioned that modern technology tended to turn human activities into mechanical processes quietly. For training, this translates into how GenAI has the potential to redefine learning for us, making it something we can maximise, automate, and quantify. If we do not take a step back, we risk allowing GenAI to redefine education in terms of outcomes and efficiency, rather than dialogue, questioning, and ethical development.

Ethical blind spots in policy and organisational culture

Although these issues are pertinent, they are hardly addressed in Colombia's AI policies. The *Roadmap for the Ethical and Sustainable Adoption of AI (MinCiencias, 2024) encourages ethical adoption, but its orientation is predominantly regulatory, focusing on the use of data, bias, and innovation*. The *CONPES 4144* (DNP, 2025) primarily positions GenAI as a contributor to productivity and growth, without providing insight on how GenAI can be incorporated ethically in learning or training spaces.

This policy silence is not coincidental. According to Coeckelbergh(2018), AI is often implemented in depoliticised contexts, where decisions regarding its function are viewed as technical or necessary rather than open to debate. In this void, organisations and students must decide what is responsible or ethical. The result is not always bad faith, but it is often characterised by confusion, inconsistency, and silence.

Establishing critical trust: A pedagogical commitment

To progress, organisations need to move beyond toolkits or guidelines. They need to be committed to creating learners who are not merely competent, but also critically and ethically equipped. This means designing training programs that:

- Discuss the potential limitations and biases of GenAI systems.
- Encourage students to ask, check, and contextualise AI-generated responses.
- Integrate real-world cases and ethical dilemmas into the learning process.
- Encourage human judgment, not only technical efficiency.

Trust in GenAI is earned, not by the tool itself but by our faith in the people who use it. Ethical AI training is all about creating professionals who realise that technology is not objective and that their task is not just to use tools but to define how tools are used.

This is of particular concern in the tech scene in Colombia, where accelerated digitalisation is not yet accompanied by profound ethical consideration. Suppose we are genuinely committed to applying GenAI to develop the workforce. In that case, we need, in addition to platforms and tools, to invest in the capacity of individuals to think with and against technology, to learn not just faster but better.

5.6 Public policy gaps and the need for coordinated action

Although Colombia's efforts to position itself as a regional leader in Artificial Intelligence are on the rise, substantial gaps remain in the policy framework, particularly regarding the role of GenAI in preparing the workforce. Though national documents such as CONPES 4144(DNP, 2025) and *Roadmap for the Ethical and Sustainable Adoption of A (MinCiencias, 2024)* articulate broad aspirations concerning responsible and beneficial AI adoption, they do not provide specific, actionable instructions for how GenAI should be incorporated in training systems for digital and technical talent.

Several interviewees, particularly from educational institutions and HR departments, observed that these policy documents are challenging to translate into actionable strategies. According to José (Educator, 2025), "*he country talks about transformation, but the actors that need to make it happen don't know how to operationalize it.*" This disconnect reflects a deeper issue: that policy often sounds good on paper. However, it is not connected with the real requirements and capabilities of training institutions, tech companies, and professionals themselves.

Fragmentation and the absence of shared governance

One of the most enduring barriers identified in this study is the lack of intersectoral coordination. While government agencies, technology companies, and academic institutions are independently exploring the adoption and implementation of GenAI, there is limited evidence of common approaches or roadmaps being developed. This

fragmentation hinders the creation of integrated learning ecosystems, making it more challenging to scale effective practices.

There is a key paradox that the tech industry, in particular, is facing. While it is often one of the first to apply GenAI tools in practice, it is rarely involved in formulating educational or policy strategies for its use. This divide isolates professionals from governance processes and constrains their role in shaping the ethical, academic, and practical criteria that should guide GenAI-based training. As Storey and Wagner(2024)note, the effective implementation of GenAI in adult education requires the active participation of all stakeholders, including those with technical and contextual knowledge.

Public investment in Colombia continues to focus on Research and Development, alongside business innovation, rather than investing in training programs and workforce development. The insufficient funding prevents qualified professionals from accessing premium training programs, which affects smaller tech businesses and workers making digital career shifts. The lack of coordination between public and private actors and limited institutional capacity, according to Sidhu et al.(2024), prevent many emerging economies from turning policy goals into practical implementation.

Establishment of a National AI workforce development strategy

A national strategy that establishes direct links between GenAI adoption and tech workforce development represents the essential solution for Colombia. The strategy requires precise cross-sectoral commitments to specify how GenAI will support talent development while ensuring inclusive practices and sustainable growth of capabilities.

It should rely in particular on the following elements:

- The government should establish national guidelines that outline ethical standards and educational approaches for GenAI use in digital talent training.
- The program should establish joint initiatives between tech firms, startups, academic institutions, and public bodies to develop training methods through collective efforts.
- The plan should provide support systems to develop the capabilities of educators, HR professionals, and tech leaders for effective GenAI implementation.
- Public investment should fund shared infrastructure along with learning platforms which companies and professionals throughout the country can access.

The main objective should not be decision centralisation but rather establishing coordinated action through shared responsibilities and priorities. The lack of alignment

in GenAI adoption will maintain its fragmented and uneven character, thus giving advantages to select groups at the expense of many others.

A strategy that unites policy goals with institutional capabilities and professional insights about tech work will enable Colombia to position itself as a leading adopter of AI in the region. As Sidhu et al. (Sidhu et al., 2024) explain, many emerging economies struggle to move from policy ambition to practical implementation due to limited institutional capacity and a lack of coordination between public and private actors. This often results in fragmented experimentation and missed opportunities for systemic impact.

Bridging this gap will require moving from rhetoric to action, from innovation debate to inclusive implementation. GenAI must be adopted as a fundamental national project that treats training, collaboration, and human development as essential pillars of sustainable digital transformation

5.7 From promise to responsibility: reflections on GenAI and the future of workforce learning

The integration of generative AI in workforce development is often presented as a question of platforms, tools, and potential future developments. However, this research has demonstrated that the true challenge is more complex and involves vision, purpose, ethics, and coordination. It concerns whether, in a world that is changing more quickly than many institutions can keep up, we are genuinely prepared to reconsider how we develop, nurture, and transform human potential.

The six themes explored in this chapter reflect different sides of that broader challenge. Some highlight structural limitations and organizational inertia. Others reveal a lack of pedagogical direction or ethical grounding. There is a promise, of course, GenAI brings clear opportunities for personalised learning and innovative approaches to training. But without coherent policies or collective frameworks, those opportunities risk remaining isolated and underused. What is clear is that GenAI is not just another new tool. It marks a shift, one that calls for imagination, intention, and shared accountability.

One of the most remarkable insights was that none of the organisations I interviewed or the companies where tech talent currently works seem to have a coherent plan or framework for using AI or GenAI in the context of upskilling or reskilling. In some instances, AI had not even come up within their Learning and Development teams. This

was not due to any sort of active pushback, but rather an absence of framework, direction, or a simple idea on how to even get started.

Despite all these gaps, I continue to hold onto hope. In my opinion, GenAI has untapped potential in democratising training. It can make learning more accessible, affordable, and tailored to the individual and their specific circumstances. It provides a real opportunity for cash-strapped companies to design training that helps people step into positions they would not have been hired for, but could flourish in with the right investment. This is not only a business advantage. It's advantageous for humanity. It removes obstacles. It paves the way to closing skill gaps and expanding opportunities for many, particularly in countries like Colombia.

This chapter highlights the need to reconceptualise GenAI's role and potential impact as more than mere automation features in technology and education. It urges us, as educators, employers, and policy makers, to pose more thoughtful questions. Not simply, *"What can this tool do?"* but rather, *"What kind of learning environments do we want to create? And who do we want to include in them?"*

The following chapter continues this reflection by offering a practical guideline. It outlines several recommendations and strategies designed to foster the timely adoption and integration of GenAI in the training of tech professionals, especially in emerging contexts like Colombia, where readiness, inclusivity, and sustainable impact are crucial.

6 Strategic Guidelines for GenAI Implementation in Tech Talent Training

This section presents the artefact of this study, a set of strategic guidelines derived from the findings of this research, in response to the third research question: *How can we utilise GenAI to bridge skill gaps in the tech workforce, particularly in emerging economies like Colombia?*

Instead of providing a general checklist of activities, these guidelines aim to integrate the practical context and evidence-informed insights derived from the analyses and literature review. They aim to support instructional designers, faculty members, and educational administrators in developing GenAI-centred training systems that are efficient, equitable, pedagogically valid, and aligned with workforce expectations.

Structure and logic

To ensure clarity and applicability, the guidelines are structured around the ADDIE model, a widely used framework in instructional design that outlines five key phases: Analysis, Design, Development, Implementation, and Evaluation. This model serves as a foundational framework in instructional design. It provides a systematic approach for creating effective and efficient training programs. (Branch, 2009)

Each phase is used here as a container to group relevant practices and decisions that organisations must consider when incorporating GenAI into their talent development strategies.

However, based on the findings of this study, it became clear that a set of cross-cutting strategic principles must also guide any attempt to integrate GenAI, critical insights that emerged consistently across both organisational and educational contexts. These are not confined to a single phase of the ADDIE model, but operate transversally, influencing every decision along the process.

Each of the five ADDIE phases will be presented as sections of the proposed guidelines in the following segments, combining a dual lens: practical recommendations grounded in empirical findings and literature, and strategic principles that ensure contextual impact and relevance to the realities of emerging economies, such as Colombia.

6.1 Analyse

Strategic analysis and alignment before implementation

The initial step in this model describes how to address the learning needs of the organisation, identify existing skill shortages, and examine the broader contexts that surround the adoption of GenAI. In the context of tech workforce development in Colombia and other emerging economies, this phase is critical, not just because of the relatively rapid pace of evolution in job roles, but also due to the numerous training systems that are still fragmented, rudimentary, or unintegrated.

The findings from this research indicate that gaps in technology capacity also coexist with a lack of readiness, alignment with organisational strategy, and fundamental concepts. Therefore, this phase must extend beyond needs assessment to become a space for reflection, planning, and stakeholder engagement.

Practical Recommendations

- **Conduct a role-based skill diagnosis using GenAI:** Organisations should begin by identifying the specific skills required for each role. GenAI tools can assist in

diagnosing competency gaps by analysing job descriptions, learner profiles, and industry benchmarks.

- **Assess organisational readiness:** Readiness is not limited to having digital infrastructure; it also involves leadership engagement, instructional capacity, and cultural openness to change.

- **Involve cross-sectoral stakeholders early in the process:** Create "Advisory Boards", where you can engage HR teams, instructors, managers, tech leads, and learners to collaboratively define the purpose and direction of the GenAI strategy.

- **Define a clear purpose for GenAI use in training:** The decision to adopt GenAI should emerge from a pedagogical goal, rather than technological enthusiasm. Clarifying the "why" helps avoid superficial deployments and increases effectiveness.

6.2 Design

Designing learning strategies with GenAI in mind

In the Design phase, the instructional strategy is conceptualised, including what learners will achieve, how learning will occur, and which resources or formats will be used. When integrating GenAI into training systems, this stage is crucial, presenting key opportunities to integrate AI-driven capabilities into the architecture from the outset, rather than as an additional element.

This study found that most organisations still trust in traditional course structures, often uploading ready-made course materials to learning management systems without customising them to the requirements of particular positions. Personalised content and learning pathways can be developed with the help of GenAI, which can also provide advanced designs, competency alignment, and collaborative frameworks.

Practical Recommendations

- **Use GenAI to co-design personalised learning paths:** GenAI models can help generate tailored roadmaps that align with learners' future goals and personal expectations.

- **Simulate occupational for validation:** GenAI can simulate how a professional in a target role would experience the training plan, identifying missing topics or misaligned learning outcomes. This approach was implemented at Nodo (EAFIT), where AI-generated role-based profiles ensured curricular coherence.

- **Involve human experts in GenAI-assisted design processes:** Regardless of whether you are using AI-generated content, human instructional designers must validate and adapt suggestions to align with pedagogical standards and stakeholders' demands.

6.3 Develop

Building AI-supported content and learning materials

In the Develop phase, the training plan created in the preceding stage is now implemented, with learning resources and activities ready for actual execution. Here is where GenAI's creativity and generative capabilities shine the most, enabling organisations to move from passive content production to active content creation driven by data.

Several participants in this study noted that they use or could use GenAI to assist in creating learning content, suggesting whole modules or generating base materials like quizzes, scripts, or slides. Nevertheless, the most effective approaches demonstrated that any output produced by AI is fundamentally deficient unless it is enhanced through oversight by a human instructor, guided by instructional logic, and subject to iterative refinement.

Practical Recommendations

- **Utilise GenAI to efficiently prototype learning materials:** GenAI tools can assist in generating draft versions of texts, activities, case studies, or visual materials, thereby accelerating the content creation process while enabling trainers to focus on making strategic adjustments.
- **Develop modular and role-specific content structures** by breaking content into learning modules tailored to distinct learner profiles or roles. GenAI can assist in generating variations of the same material adapted to different levels of complexity or formats.
- **Ensure human-in-the-loop validation of all outputs.**
  AI-generated content should be reviewed and improved by instructors or instructional designers to ensure alignment with learning outcomes and pedagogical quality.Remember, one of the risks of GenAI-generated content is its trustworthiness.

6.4 Implement

Delivering GenAI-supported training

The implementation phase involves the deployment of the training program and its delivery to learners. In GenAI-enabled systems, this is where strategy comes to life, where the set

of learning materials, systems, and experiences crafted in the previous phases is deployed in real-world settings.

This research revealed that many organisations continue to implement GenAI in fragmented or exploratory ways, through labs, pilots, or isolated tools. However, integrating GenAI into existing learning ecosystems and aligning it with workflows, roles, and digital platforms is also very promising.

Practical Recommendations

- **Pilot GenAI-based training in controlled environments**: Begin with limited deployments as a test, targeting specific teams, roles, or skills, to evaluate impact and usability, and lower the risks. Feasible ideas are AI Labs or sandbox environments, which allow safe experimentation before scaling.
- **Provide onboarding and support for both learners and facilitators**: Introducing GenAI into learning processes often necessitates cultural and behavioural adjustments. Offer initial guidance, explain the role of AI, and ensure support channels are in place.

- **Deploy GenAI-powered learning assistants or chatbots:** One of the most impactful and scalable practices identified is the use of GenAI agents that serve as on-demand learning tutors**.** These chatbots can answer questions, guide learners, recommend next steps, or monitor progress, allowing for an expanded reach. As Camilo (L&D, 2025) expressed, "*Using a GenAI assistant means we don't need 10 people checking who's behind on their learning. The system can follow up and suggest what's next"*.

  This can significantly reduce instructional costs while offering scalable, personalised support, especially valuable in resource-constrained contexts like those found in Colombia.

6.5 Evaluate

Measuring the impact and refining GenAI

The Evaluation stage of the model represents the last step. Yet, it remains the least developed and most overlooked component when it comes to integrating GenAI in training, learning, and development strategies. Based on the findings of this study, none of the companies involved have yet considered how to evaluate GenAI-based training. In many cases, there is no plan, no framework, and no clarity on what success would look like when implementing AI in talent development.

The absence of evaluative thinking suggests that GenAI is primarily viewed as a content generation tool rather than a strategic component that requires monitoring and accountability. This phase represents the most significant gap in current strategies,

offering organisations a major opportunity to develop scalable learning ecosystems that are responsible and focused on achieving valuable outcomes.

Some participants emphasised that organisations need to establish strong evaluation systems to prove their return on investment (ROI). According to Jessica (Scrum Master, 2025), organisations need to demonstrate concrete results because this proves essential for both initiative expansion and internal stakeholder support: *"If we don't know whether this is working or not, whether it's saving time or improving performance, how can we justify expanding it?"*.

The perspective demonstrates that evaluation extends beyond learning analytics; it's about evidence-based decision-making and resource allocation based on the redistribution of organisational priorities.

Practical Recommendations

- **Use GenAI to support learning analytics and feedback analysis:** GenAI can help analyse survey responses, learner progress, and usage patterns to identify strengths, gaps, and potential areas of disengagement. This supports real-time improvements and long-term process tracking.

- **Evaluate outcomes and real-world impact:** Go beyond tracking completions, evaluate whether training improves performance, closes specific skill gaps, or supports career advancement. This aligns the use of GenAI with strategic business and workforce goals.

- **Use humans to interpret AI-driven insights:** AI can process data at scale, but human judgment is crucial for understanding context, validating insights, and guiding improvements. Evaluation should be a process built with peers

- **Assess ethical and social effects of GenAI:** Explore how learners perceive the role of AI in their development. Do they trust the system? Do they feel supported or replaced? Include indicators related to transparency and inclusion.

6.6 Transversal Strategic Pillars

In addition to the guidelines presented, which utilise the ADDIE model as their structural foundation, this case study has identified five strategic principles that apply to the entire continuum of the design and deployment process of GenAI-enabled training strategies. These principles, derived from qualitative analysis and a literature review, transcend the

phases of the process; they operate transversely, shaping the effectiveness, relevance, and equity of GenAI adoption across diverse contexts.

The principles reflect essential tensions and possibilities encountered primarily in tech organisations and also in higher education institutions in Colombia, ranging from fragmented implementation and unclear goals to promising yet unstructured innovation efforts. These principles aim to assist policymakers, instructors, and curriculum developers in proactively and systematically maintaining a long-term, ethical, and inclusive vision as they scale GenAI-based learning strategies.

1. **Intentional use and readiness**
   GenAI should not be adopted reactively or opportunistically, but intentionally, anchored in a well-defined learning strategy supported by leadership and institutional capacity.

2. **Learning Integrity and Ethics**
   Any AI-generated content must be critically reviewed to ensure it aligns with comprehensive pedagogical practices, ethical standards, and meaningful learning goals.

3. **Personalised learning with purpose**
   GenAI should be utilised to personalise learning pathways based on actual occupational needs, enabling learners to pursue and plan their professional development.

4. **Active and reflective learning**
   The environments in which learning occurs must foster safe experimentation, promote learner autonomy, and encourage critical engagement with AI tools, rather than merely facilitate passive content consumption.

5. **Equity and access in practice**
   GenAI solutions should be accessible and affordable, especially in low-resource contexts. They must be designed to reduce, not deepen, existing inequalities in skill access and opportunity.

6.7 Conclusions

The objective of this study was to investigate the application of Generative AI (GenAI) in facilitating upskilling and reskilling within the tech sector in Colombia. The study employed 13 expert interviews from the education, human resources, and technology industries, exploring the adoption of this emerging technology, the challenges it presents, the awareness of individuals regarding its utilisation, and potential strategies for optimising its effectiveness.

The summarised findings are presented in four broad themes: opportunities, challenges, perceptions, and strategies.

Opportunities

GenAI presents actual opportunities to enhance professional learning, development, and career growth for individuals. The technology enables organisations to create individualised training routes, which accelerate professional development and decrease operational expenses. Multiple organisations currently employ GenAI to develop career development strategies and provide learning recommendations and AI-based support to their learners. GenAI transforms into an organisational tool when organisations define specific goals, enabling them to develop advanced training methods for their teams. The opportunities are evident to enhance training, making it more efficient, personalised and scalable.

Challenges

The adoption of GenAI remains in its initial phase despite its vast potential. Most organisations try out new approaches, but only a few have established concrete plans and strategies. The main obstacles to adoption stem from organisational and educational challenges rather than technical difficulties. The current lack of coordination, combined with limited internal skills and insufficient knowledge about GenAI learning support, creates barriers. Public policies remain unclear and separate from the operational realities of training institutions and technology companies. The successful implementation of GenAI at scale requires more than just tools, as it requires structured leadership and a unified vision.

Perceptions

The participants expressed diverse opinions about GenAI. People are both interested in its capabilities and concerned about its potential risks, which include replacing human trainers, losing control of content, and excessive dependence on automated tools. There is also a strong call for ethical contemplation together with critical thinking. GenAI requires trust from users, which is not given for granted; it must be earned. This means we must use it carefully, ensuring it supports learning rather than replacing it, and keeping humans always in the loop.

Strategies

Organisations have implemented various innovative approaches to integrate GenAI into their operations. They utilise GenAI to develop training content through collaborative efforts between AI systems and educators, while establishing AI Labs for experimental purposes and applying GenAI to identify learning requirements based on specific roles. The research cases serve as the foundation for this thesis, which aims to develop operational guidelines that enable others to replicate similar approaches. The guidelines follow the ADDIE model (Analyse, Design, Develop, Implement, Evaluate), and are supported by key principles: intention and readiness, integrity and ethics, learning with purpose, equity and access, and

active learning. These strategies aim to help educational institutions and companies in transitioning GenAI from an experimental concept to a sustainable, long-term solution.

GenAI as a lever for inclusion, employability, and systemic change
This research demonstrates that GenAI can, when implemented strategically, go beyond optimising learning to overcoming critical workforce development issues. GenAI provides the opportunity to scale training programs and streamline preparation for the digital economy in Colombia, a country hindered by economic inequality, a digital divide, and fragmented institutions.

These findings directly support the advancement of the sustainability agenda:

- **SDG 4 (Quality Education):** GenAI can help expand access to personalised, relevant, and affordable learning opportunities, especially for young people and underserved communities.
- **SDG 8 (Decent Work and Economic Growth):** By accelerating upskilling and reskilling, organisations can close talent gaps more efficiently and create new professional pathways.
- **SDG 10 (Reduced Inequalities):** If implemented with a focus on inclusion, GenAI can reduce systemic disparities in access to high-quality training and digital employment opportunities.

This thesis contributes to the discussion on how technology, particularly GenAI, can be leveraged for meaningful social change, developmental impact, and inclusive economic growth. It emphasises that innovation in workforce training transcends tools; it is about purpose, systems, and people.

6.8 Limitations and Future Research

The research provides essential knowledge about the implementation of GenAI workforce training in Colombia; however, several important limitations exist:

- The research design employs qualitative methods with 13 participants, which provides detailed findings but limits the ability to generalise the results. Future research should employ mixed-methods approaches in combination with large-scale surveys to validate the identified themes.

- The research gathered insights from tech professionals, L&D leaders, and educators. Still, it failed to include perspectives from policymakers and learners who do not work in professional fields, such as unemployed youth and early-career

learners. The inclusion of these voices would enhance the comprehension of GenAI experiences and perceptions among the broader population, especially those who benefit from the technology.

- Finally, although ethical considerations were discussed, they were not the central focus of this study. Due to the importance of trust, transparency, and agency in learning interventions, further research is needed to understand how ethics can be participatively designed with learners and practitioners in AI-driven educational settings.

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8 Appendices

8.1 Appendix A

A. Literature Review Process

The Literature Review was conducted according to the following process:

8.1.1.1 Search Strategy

Sources

The sources used to collect the information were:

- Google Scholar: a web search engine from Google that specifically searches scholarly literature and academic resources.(University, n.d.)
- ERIC is a comprehensive internet-based bibliographic and full-text database of education research and information.(*ERIC FAQ - General*, n.d.)
- IEEE Xplore: Is a digital platform for discovery and access to scientific and technical content published by the IEEE.(*About IEEE Xplore*, n.d.)

Types of publications:

The publications included in this Literature Review are of two types:

- Academic literature: Publications subject to a rigorous peer-review process, such as articles in indexed journals, book chapters and doctoral theses, produced by researchers affiliated with universities and research centres.
- Business literature: Reports and studies produced by companies and organisations, which may or may not be subject to peer review, offering insights into private and/or public sector practices and challenges.

Main Themes:

All the publications collected had to be related to the following themes:

- T1: Socio-economic factors influencing AI Adoption
- T2: Generative AI in adult learning
- T3: Generative AI in workforce development in emerging countries
- T4: The use of AI in upskilling and reskilling the workforce.

Keywords:

In order to develop the Search Strings, I organised the keywords or terms into three groups:

- G1: Terms related to AI → Generative AI OR AI OR GenAI OR Artificial Intelligence OR AI Adoption
- G2: Terms related to Workforce development → Workforce OR Skill Development OR Workforce Development OR Upskilling OR Reskilling
- G3: Terms related to Economic factors → Emerging Markets OR Developed Markets OR Socio-Economic factors OR Colombia

I created three search strings by combining the terms from each group. The general search string was applied to both the IEEE Xplore and ERIC databases. To generate fewer but more targeted results in Google Scholar, I developed two alternative search strings that incorporated all the group terms.

  - General Search String (IEEE + ERIC): (Generative AI OR AI OR GenAI OR Artificial Intelligence OR AI Adoption) AND (Workforce OR Skill Development OR Workforce Development OR Upskilling OR Reskilling) AND (Emerging Markets OR Socio-Economic factors OR Colombia)
  - Google Scholar:
    - AI AND ("Upskilling" OR Workforce OR Reskilling OR Skill)
    - AI AND Emerging Economies

8.1.1.2 Inclusion Criteria

- Articles published between 2020 and 2024
- Articles written in English and Spanish
- Publications that are directly and deeply related to one of the 5 defined thematic areas.
- Peer-reviewed articles
- Publications that are directly and deeply related to at least one of the three research questions.

8.1.1.3 Exclusion Criteria

- Articles that were published before 2020
- Articles written in languages other than English and Spanish.
- Any publications that focus on the impact of AI on innovation in areas other than adult learning and education.
- Articles that are not Peer-Reviewed
- Publications that are not deeply related to one of the 5 thematic areas defined.
- Publications that do not directly and deeply relate to at least one of the three research questions.

8.1.1.4 Classification and evaluation of publications

After applying the search strings in the databases, a total of 1,598 were retrieved. You can observe the details in Table X.

Database	Search String	Hits
Google Scholar	AI AND (Upskilling OR Workforce OR Reskilling OR Skill)	26
	AI AND Emerging Economies	2
ERIC	General Search String	1410
IEEE Explore	General Search String	160

Table 7. Database Search Hits

These articles went through a careful screening and selection process to narrow down the sample to only the most relevant ones. First, I filtered those articles based on their availability and publication date, ensuring they were published after 2020, thereby applying two of the inclusion and exclusion criteria. Then, I re-read the titles and abstracts of those articles, verifying which ones best fit the focus of this research. The outcome of this Screening process was 46 articles.

Then, I classified each article into one of the five thematic areas mentioned above, assessed its relevance to the research questions, and applied the remaining inclusion and exclusion criteria. This process is illustrated in Figure X. Finally, 10 articles have been selected for review and detailed analysis, as shown in Table Y.

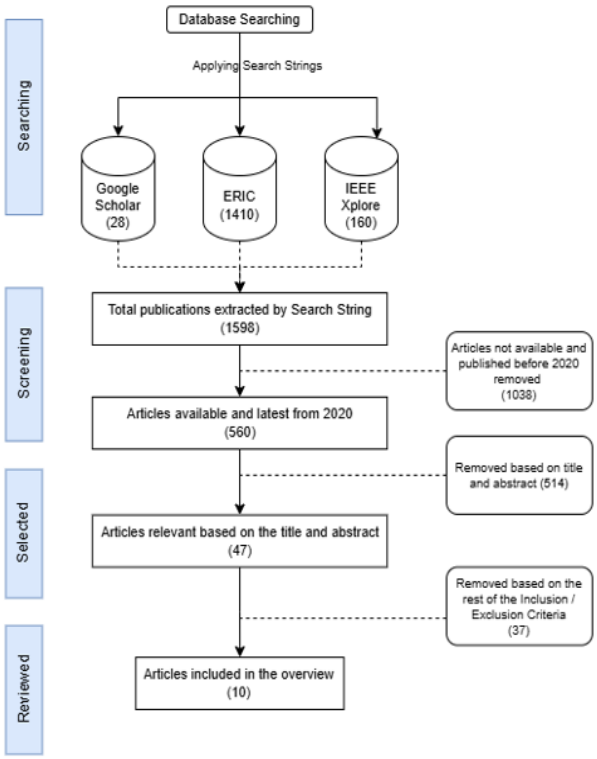


Figure 8. Research Process

Publications Title	Theme	Data Source	RQ
AI and Workforce Development: A Comparative Analysis of Skill Gaps and Training Needs in Emerging Economies:	Generative AI in workforce development in emerging countries	Google Scholar	RQ2
Bridging the Skill Gap in IT by Using Generative AI: Utilizing Opportunities and Overcoming Challenges	Generative AI in workforce development in developed countries	Google Scholar	RQ1
Adopción de Tecnologías de Inteligencia Artificial: Un Estudio para las Empresas en Colombia	Generative AI in workforce development in emerging countries	Google Scholar	RQ2
Integrating Artificial Intelligence (AI) into Adult Education: Opportunities, Challenges, and Future Directions	Generative AI in adult learning	ERIC	RQ1
Artificial Intelligence in Corporate Learning and Development: Current Trends and Future Possibilities	The use of AI in upskilling and reskilling the workforce	IEEE Xplore	RQ1
The Impact of Artificial Intelligence on Workers' Skills: Upskilling and Reskilling in Organisations	The use of AI in upskilling and reskilling the workforce	Google Scholar	RQ3
AI Adoption in Universities in Emerging Economies: Prospects, Challenges and Recommendations	Socio-economic factors influencing AI Adoption	Google Scholar	RQ1
Artificial Intelligence in the Industry 4.0, and Its Impact on Poverty, Innovation, Infrastructure Development, and the Sustainable Development Goals: Lessons from Emerging Economies?	Socio-economic factors influencing AI Adoption	Google Scholar	RQ3
Artificial intelligence in developing countries: The impact of generative artificial intelligence (AI) technologies for development	Generative AI in workforce development in emerging countries	Google Scholar	RQ2

Table 9. Selected publications.

8.2 Appendix B - Interview Guide

Socio-demographic Information

- Name (optional; depending on confidentiality preference)
- Gender (optional)
- Age group
- Nationality / Country of operation
- Role / Job Title
- Industry / Type of Organisation
- Years of experience
- Familiarity level with Generative AI (GenAI)

Block 1: Introduction and Contextual Understanding

All Interviewees

A. Could you briefly describe your role and responsibilities within your organization?
    a. Could you briefly describe your role and the number of years you've been working in this field?
B. How would you describe your current knowledge or involvement with AI and generative AI?

Block 2: Learning Challenges and Current Upskilling/Reskilling Practices

Tech Professionals

A. From your experience, what are the main challenges you face when learning new technical or soft skills?
B. How do you typically learn new skills at your workplace, and how effective are these methods?
C. What kind of tools or support do you think are currently missing or could significantly improve your company's approach to skill development?

HR Professionals

A. What key challenges does your organization face in identifying and addressing skills gaps among employees?
B. Could you briefly describe your current methods for upskilling and reskilling, along with their primary limitations?
C. Have you considered integrating GenAI or other advanced AI technologies to enhance your training strategies? Why or why not?

Educational Institution Professionals

A. What are the primary challenges your institution faces when preparing tech professionals for current industry demands?
B. Could you highlight your most effective current approaches or methods for reskilling and upskilling tech professionals?
C. What current gaps or limitations exist in your training approach that you think could potentially be addressed by Generative AI?

Block 3: GenAI adoption and Use

Tech Professionals

A. What are the key skills currently required in your role, and how have they changed with AI advancements?
B. Can you provide examples of how generative AI is being used or might be used in your organisation to enhance skill development?

HR Professionals

A. How is your organisation integrating Generative AI into employee training programs?
    a. Why are you not integrating GenAI?
B. What criteria do you use when selecting AI-driven tools for workforce development?

Educational Institution Professionals

A. In what ways have your programs adapted to integrate GenAI technologies?
    a. Why are you not using GenAI in your programs?
B. Could you describe a specific case or initiative where GenAI improved your learning outcomes?

Block 4: Challenges and Barriers

All Interviewees

A. What significant challenges or barriers have you identified in adopting Generative AI for workforce development in Colombia?
B. How do factors such as infrastructure, regulation, or ethics influence the implementation of Generative AI solutions?
C. From your experience, what do you see as the main strengths or advantages of your current context (country or sector) that could facilitate effective adoption of technologies like Generative AI for professional use?

D. What specific initiatives or practices would you seek to incorporate into your own country or organisation to improve AI-driven training?

Block 5: Opportunities, Best Practices and Recommendations

All Interviewees

A. In your view, how can Generative AI transform workforce development programs to become more effective?
B. What kind of support (e.g., policy changes, partnerships, resources) would be necessary to leverage AI more effectively for training?

Block 6: Ethical Concerns

All Interviewees

A. What ethical considerations have arisen or might arise from using Generative AI in workforce training?

Block 7: Future Perspectives

All Interviewees

A. Can you propose any specific strategic recommendations or guidelines to optimize and/or effectively apply GenAI implementation for workforce training?

Closing: Open Comments

All Interviewees

A. Do you have any additional insights or thoughts related to this topic?

8.3 Appendix C – Letter of consent

Faculty of Technology and Society
Data collection consent
Master Thesis Tittle: Bridging the Skill Gap: The Role of AI in Upskilling and Reskilling the Tech Industry

Letter of Consent for interview research

You are invited to participate in a research study that explores the role of Generative Artificial Intelligence (GenAI) in workforce development, specifically focusing on how this technology can support upskilling and reskilling efforts in Colombia and Sweden. This study is conducted as part of my Master's thesis in the Master's program in Computer Science and Innovation for Change at Malmö University.

The main objective of this research is to understand how IT companies and higher education institutions in these two distinct economic contexts are adopting GenAI to address skill shortages, and to identify challenges, opportunities, and best practices.

The interview will last approximately 60–85 minutes and will be conducted by me, the researcher. With your consent, the session will be audio-recorded for transcription and analysis.

Your participation is voluntary, and all information shared will be treated with the highest level of confidentiality. Your identity will be anonymized in the final report, and no personal identifiable information will be disclosed. All recordings and transcripts will be stored securely, accessible only to the researcher, and deleted upon final approval of the thesis.

The data collected will be used exclusively for academic purposes. Research findings may be published in academic forums, but no commercial use will be made of the data. A summary of the findings and a copy of the final thesis will be shared with all participants upon request.

Your participation is entirely voluntary. You may decline to answer any questions or withdraw from the study at any time without providing a reason. You may also request to review the transcript of your interview and suggest changes or clarifications.

If you have any questions or concerns, feel free to contact me or my supervisor.

Kind regards,

Juan David Ramirez (Student)
juan.ramirez.mau@gmail.com

Katarzyna Gruszka, PhD(Supervisor)
katarzyna.gruszka@mau.se

8.4 Appendix D - Codebook

Code	Parent Code	Related RQs
Contextual Enablers	AI Readiness Requirements	RQ1
Curriculum and Institutional Capacity	AI Readiness Requirements	RQ1
Digital Literacy	AI Readiness Requirements	RQ1
Infrastructure	AI Readiness Requirements	RQ1
Perceived Contextual Limitations	AI Readiness Requirements	RQ1
Policy and Ecosystem Recommendations	AI Readiness Requirements	RQ1
Cost or Resource Constraints	General GenAI Adoption & Implementation Challenges	RQ1
Limited Skilled Talent	General GenAI Adoption & Implementation Challenges	RQ1
Organizational Resistance	General GenAI Adoption & Implementation Challenges	RQ1
Regulatory and Policy Gaps	General GenAI Adoption & Implementation Challenges	RQ1
Misconceptions and Awareness Gaps about GenAI	General GenAI Adoption & Implementation Challenges	RQ1, RQ2
Knowledge Management Potential	General GenAI Adoption & Implementation Opportunities	RQ1
Scalability Potential	General GenAI Adoption & Implementation Opportunities	RQ1
Visionary GenAI Applications	General GenAI Adoption & Implementation Opportunities	RQ1
GenAI Applications in Training	GenAI Adoption & Implementation Opportunities in Training and Talent Development	RQ1
Innovation in Training	GenAI Adoption & Implementation Opportunities in Training and Talent Development	RQ1
Opportunities for Personalization	GenAI Adoption & Implementation Opportunities in Training and Talent Development	RQ1
Learning Support	GenAI Adoption & Implementation Opportunities in Training and Talent Development	RQ1
Absence of GenAI in Training	GenAI Adoption & Implementation Challenges in Training and Talent Development	RQ1

Organizational Readiness for GenAI in Training	GenAI Adoption & Implementation Challenges in Training and Talent Development	RQ1
Corporate Training Strategies	Learning & Training Approaches	RQ1, RQ3
Learning Strategy Orientation	Learning & Training Approaches	RQ1, RQ3
Learning Practices	Learning & Training Approaches	RQ1, RQ3
AI-Influenced Skills and Competency Shifts	Learning & Workforce Development Needs	RQ1, RQ3
Learning & Training Challenges	Learning & Workforce Development Needs	RQ1, RQ3
Training Delivery Constraints	Learning & Workforce Development Needs	RQ1, RQ3
Alignment with Local Context	Stakeholder Perceptions	RQ2
Equity and Inclusion Views	Stakeholder Perceptions	RQ2
Ethical Concerns	Stakeholder Perceptions	RQ2
Perceived Efficacy of GenAI	Stakeholder Perceptions	RQ2
Perceived Scalability	Stakeholder Perceptions	RQ2
Perceptions of Colombia's Strategic Positioning	Stakeholder Perceptions	RQ2
Perceptions of National AI Governance	Stakeholder Perceptions	RQ2
Perceptions of National Digital Readiness	Stakeholder Perceptions	RQ2
Positive Perceptions of GenAI Potential	Stakeholder Perceptions	RQ2
Trust in AI for Training	Stakeholder Perceptions	RQ2
Vision for Strategic AI Integration	Stakeholder Perceptions	RQ2
Community-Based Innovation	Strategic Solutions for GenAI in Talent Development	RQ3
Partnership Opportunities	Strategic Solutions for GenAI in Talent Development	RQ3
Proposed Strategies	Strategic Solutions for GenAI in Talent Development	RQ3
Scalability and Sustainability Models	Strategic Solutions for GenAI in Talent Development	RQ3
Skills-to-Employment Alignment	Strategic Solutions for GenAI in Talent Development	RQ3
Successful Use Cases / Pilots	Strategic Solutions for GenAI in Talent Development	RQ3